Universidade do Minho
Escola de Engenharia

Luís Paulo Peixoto dos Santos

Quantum Machine Learning

Habilitation in Computer Science
Curricular Unit: report

January 2026

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Universidade do Minho, Braga, January 2026

Luís Paulo Peixoto dos Santos

Resumo

Este texto constitui o relatório submetido à Universidade do Minho para preenchimento parcial dos requisitos para realização de Provas de Agregação no ramo do conhecimento de Informática nos termos do número 2 do artigo 8º do DL 239/2007, de 19 de Junho, cuja alínea b) menciona *"... um relatório sobre uma unidade curricular, grupo de unidades curriculares, ou ciclo de estudos, no âmbito do ramo do conhecimento ou especialidade em que são prestadas as provas"*.

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A Unidade Curricular é descrita em detalhe, apresentando-se também uma análise dos resultados do seu funcionamento nos anos lectivos de 2021/22 a 2023/24.

Palavras-chave Unidade Curricular, Agregação, Computação Quântica, Aprendizagem Máquina

Abstract

This document intends to partially fulfill the requirements of University of Minho to grant the academic title of Habilitation in Computer Science.

It describes a course on Quantum Machine Learning offered in the context of the Master in Physics Engineering, specialization in Physics of Information. The course is characterized using the European Competence Framework in Quantum Technologies. It is then described in detail, including an analysis of the results achieved from 2021/22 to 2023/24.

Keywords Course, Habilitation, Quantum Machine Learning

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Acrónimos

ACM Association for Computing Machinery.

ARS Audience Response System.

CML Classical Machine Learning.

CU Curricular Unit.

DMN Devices, Microsystems and Nanotechnologies.

DS Data Science.

ECFQT European Competence Framework in Quantum Technologies.

ECTS European Credit Transfer System.

LEPhys BsC in Physical Engineering.

MEPhys Master in Physical Engineering.

ML Machine Learning.

PI Physics of Information.

QAOA Quantum Approximate Optimization Algorithm.

qBLAS quantum Basic Linear Algebra Subprograms.

QC Quantum Computing.

QDS Quantum Data Science.

QML Quantum Machine Learning.

QML Quantum Machine Learning.

QT Quantum Technology.

QUBO Quadratic Unconstrained Binary Optimization.

QUCe Questionnaire for **CU** assessment - student variant.

SIGAQ Quality Assurance Internal System – Universidade do Minho.

SVM Support Vector Machine.

Glossário

European Competence Framework in Quantum Technologies Mapping of skills and knowledge in Quantum Technologies. A tool that facilitates the design and comparison of education projects within the context of Quantum Technologies. Promotes the characterization of the profile of graduated students and professionals within this area of knowledge. Developed by the project Quantum Technology Education, within the context of the Quantum Flagship, sponsored by the European Union.

Qiskit Quantum computing open source software development kit (link).

Quantum Computing Utilization of phenomena prescribed by quantum mechanics for information processing.

Quantum Machine Learning Utilization of quantum computing to achieve the goals associated with Classical Machine Learning.

Chapter 1

Introduction

This document constitutes the characterisation report of a **Curricular Unit (CU)**, as required by Decree-Law DL 239/2007, for the purpose of undertaking the Habilitation Examinations in the field of Computer Science.

1.1 Object and motivation

The **CU** described in this report is entitled **Quantum Machine Learning (QML)**. This **CU** is part of the study plan of the **Master in Physical Engineering (MEPhys)**, specialisation in **Physics of Information**, and is offered in the second semester of the first year. It carries a workload of 5 **ECTS**, with a teaching component of four hours per week, divided into a two-hour theoretical lecture and a two-hour theoretical–practical class.

The **MEPhys** had its first edition in the 2021/22 academic year. It resulted from the restructuring process of the Integrated Master's Degree in Physical Engineering, which led to the creation of first- and second-cycle programmes in the area of Physical Engineering. At the time of writing this report (Summer 2024), this **CU** has completed three editions. It has one particular feature: its designation in the **MEPhys** study plan is **Quantum Data Science**. However, it is my conviction that this is not the most appropriate designation. In fact, the objectives, competences and syllabus that are defined are much more closely related to **Machine Learning** than to **Data Science**. Consequently, this report will use the designation **Quantum Machine Learning**, except when explicitly referring to the current study plan. It is my intention, at the time of the next request for reaccreditation of the programme, to request this change of designation from the Commission responsible for instructing the process.

Quantum Machine Learning is an emerging area within the broader domain of Quantum Computing, which itself is relatively young. It is a recent field undergoing constant change, with a high rate of publication of new results. It is also an area whose potential impact on science, technology and society has not yet been established. It is still unclear whether **QML** will result in any practical advantage over **Classical Machine Learning (CML)**, something that is commonly referred to as the classical–quantum separation.

The preparation of a **CU** in this scenario, characterised by innovation and challenges, requires a definition of **QML** that, in the context of this report, allows the subject matter to be clarified, delimited and motivated. In this context, we understand Machine Learning as the study, development and use of algorithms that infer models from training data and generalise to new data, performing tasks without explicit instructions. We delimit **QML** to the use of quantum resources to enhance **CML**, that is, to obtain some computational advantage in pursuing the objectives associated with the latter. The topics addressed in this **CU** concern the processing of classical data using quantum resources, with particular emphasis on supervised learning.

QML is one of the application areas of Quantum Computing that is regarded as a candidate for demonstrating the aforementioned quantum advantage. This advantage may eventually manifest itself in several dimensions, such as computational complexity, trainability, data volume, efficient exploration of hyper-dimensional spaces (*feature spaces*), and so on. **QML** is still in its infancy, with many challenges to overcome and credentials yet to be demonstrated. Nevertheless, its potential to revolutionise machine learning is significant and cannot, given the state of knowledge in 2024, be ignored in the context of a Master's degree in Physical Engineering, specialising in Information Physics.

1.2 Structure

The next chapter begins by framing this training proposal within the recommendations of the **Association for Computing Machinery (ACM)** Computer Science Curricula 2023, followed by a description of several curricular units focused on the same area of knowledge offered by other national and international higher education institutions.

Chapter 3 presents the **European Competence Framework in Quantum Technologies (ECFQT)**, which is used in Section 3.2 to characterise the qualification profile and levels of proficiency in **Quantum Computing** of a Master's graduate in Physical Engineering, specialising in Information Physics, at the University of Minho.

Chapter 4 describes the **Curricular Unit Quantum Machine Learning**, which is the object of this report, and presents some results from the three editions held in the academic years 2021/22, 2022/23 and 2023/24.

Finally, the report concludes with some final considerations presented in Chapter 5.

Chapter 2

Educational Offer at other Institutions

This chapter begins by situating the field of **Quantum Machine Learning (QML)** within the recommendations of the **Association for Computing Machinery (ACM)** Computing Curricula. Subsequently, a synopsis of the educational offer in this same area at international and national levels is presented.

2.1 ACM Computing Curricula

Quantum computing is classified as an *emerging technology* in the **ACM** Computing Curricula 2020 [4] (see Section 8.1.3, p. 88). This document does not include any recommendations regarding education in this area, merely noting that this framework may change in future curricular recommendations and acknowledging the importance attributed to it by major consultancy firms such as Accenture and Deloitte. No reference is made to the field of **Quantum Machine Learning**.

The situation changes significantly with the publication of the **ACM** Computer Science Curricula 2023 [3], in January 2024. Education in quantum computing is explicitly included in 4 of the 17 knowledge areas established in this document. These knowledge areas are: "Algorithmic Foundations" (elective), "Architecture and Organisation" (core), "Programming Languages Foundations" (core), and "Specialised Platform Development" (elective). Quantum mechanics is also mentioned in the specification of the "Networking" and "Security" areas, but in the context of (Physical) Information Theory.

The most relevant proposed syllabus contents are listed in Table 1.

Quantum Computing – Syllabus Contents

(1)	fundamentals of quantum mechanics – *qubit*, superposition, entanglement, unitary evolution, measurement, no-cloning
(2)	linear algebra – Hilbert spaces, states, operators, tensor product
(3)	quantum circuits and *gates*
(4)	algorithms – Deutsch–Jozsa, Shor and Grover
(5)	implementations – physical implementation of *qubits*, the quantum processing unit (QPU), noise and error mitigation

Table 1: Quantum Computing – syllabus contents recommended by the **ACM** Computer Science Curricula 2023 [3]

The **ACM** Computer Science Curricula 2023 [3], similarly to the **ACM** Computing Curricula 2020 [4], does not include any reference to **Quantum Machine Learning**. This is a very recent and rapidly evolving field, with very limited educational provision at first- and second-cycle higher education levels, thereby confirming the innovative nature of the curricular unit presented in this report.

2.2 International Context

Educational provision by higher education institutions in the field of **Quantum Machine Learning** is still scarce. Below, we identify some curricular units focused on this topic. We begin, however, the analysis of the international landscape with provision in the field of **Quantum Computing**, which is much more widely represented.

Educational offers in the area of **Quantum Computing** can be found at a large number of institutions around the world; Table 2 presents a representative sample of the international landscape. The website of the QTEdu – Quantum Technology Education initiative includes an extensive list of educational offers (and other resources) at the higher education level in Europe.

Quantum Computing

University	Curricular Unit / Degree	Region
Oxford University	Quantum Processes and Computation	United Kingdom
University of Surrey	Applied Quantum Computing (MSc degree)	
University of Warwick	Quantum Computing	
Sorbonne Université	Quantum Information (MsC degree)	Europe
University of Helsinki	Introduction to Quantum Computation	
Technical University of Munich	Quantum Science & Technology (MSc degree)	
Universiteit Leiden	Quantum Algorithms	
Università di Verona	Quantum Computing	
University of Amsterdam	Quantum Computer Science (MSc degree)	
California Institute of Technology	Quantum Computation	America
Johns Hopkins University	Quantum Computation	
University of Harvard	Quantum Computation and Complexity	
Cornell University	Quantum Information Processing	
Massachussets IT	Quantum Computing Fundamentals	
University of Western Australia	Quantum Computing (major)	Australia
National University of Singapore	Quantum Information and Computation	Asia
Indian IT Madras	Quantum Computing	

Table 2: Higher Education offer in Quantum Computing – international sample

It is possible to identify some characteristics shared by the majority of these educational initiatives:

- their syllabi include fundamental concepts associated with quantum computing, such as the notion of the *qubit*, superposition, entanglement, unitary evolution, quantum circuits and quantum algorithms—namely algorithms suited to fault-tolerant quantum systems, such as Grover and Shor. They also generally address concepts and techniques related to quantum information and communication, such as information theory, teleportation and quantum protocols. The syllabi are therefore largely aligned with those proposed in the **ACM** Computer Science Curricula 2024 [3] (see Section 2.1, Table 1);

- they are mostly offered by the physics departments and include concepts from quantum mechanics, showing that a causal decoupling between this discipline and the quantum computing paradigm has not yet been achieved;

- they are predominantly offered at a relatively advanced level of higher education, namely at a level which in Europe corresponds to the second cycle.

The educational provision identified in this area (see Table 2 for a representative sample) consists essentially of curricular units integrated into programmes oriented towards Physics and/or Computer Science. However, there are already some programmes fully or partially oriented towards **Quantum Computing**. This is the case for offerings from the *University of Surrey*, *Sorbonne Université*, *University of Amsterdam* and the *Technical University of Munich*, which offer second-cycle degrees in this field. The *University of Western Australia* enables the completion of a *major* at second-cycle level.

Educational provision in **Quantum Machine Learning (QML)** is far more limited, due to the specificity of the topic and its recent emergence. In fact, only in the last eight years have practical approaches to this problem begun to appear, and the current pace of innovation is frenetic.

Some of the curricular units listed in Table 2 include brief references to **QML**, namely the offerings from the Università di Verona and the Indian Institute of Technology Madras. The information available for the former is insufficient to characterise the corresponding offer. With regard to the latter, it addresses Quantum Variational Classifiers and quantum Support Vector Machines.

Table 3 lists some curricular units in the field of **Quantum Machine Learning (QML)** offered by foreign higher education institutions.

Quantum Machine Learning

	University	Curricular Unit	Cycle	Country
(1)	University of Maryland	Quantum Machine Learning	1^{st}	U.S.A.
(2)	Università Ca' Foscari Venezia	Quantum Machine Learning	2^{nd}	Italy
(3)	University College Dublin	Quantum Machine Learning	2^{nd}	Ireland
(4)	University of Oslo	Quantum Computing and Quantum Machine Learning	2^{nd}	Norway
(5)	Halmstad University	Quantum Machine Learning	3^{rd}	Sweden

Table 3: Higher Education offers in Quantum Machine Learning – international sample

Skills

Description	(1)	(2)	(3)	(4)	(5)
Utilization of **QML** tools and algorithms			•	•	
Comparison of quantum and classical machine learning			•	•	
Assessment of potential performance gains				•	
Challenges and opportunities			•		

Table 4: Skills in Quantum Machine Learning – international sample (labels (?) refer to the first column of table 3)

Syllabus

Description	(1)	(2)	(3)	(4)	(5)
Data Representation			•	•	•
Tools for development and test			•		
Performance characterization			•		
Quantum Variational Algorithms		•	•	•	•
Quantum kernels and feature maps		•	•	•	•
Algorithms based on quantum kernels		•	•	•	•
Quantum generative models		•			
Quantum reinforcement learning					•
Applications		•			•

Table 5: Syllabus in Quantum Machine Learning – international sample (labels (?) refer to the first column of table 3)

Recommended Bibliography

Description	(1)	(2)	(3)	(4)	(5)
Machine Learning with Quantum Computers Maria Schuld and Francesco Petruccione, 2021			•	•	
Quantum Machine Learning: An Applied Approach Santani Ganguly, 2021			•		

Table 6: Recommended bibliography in Quantum Machine Learning – international sample (labels (?) refer to the first column of table 3)

This sample is representative of the three cycles of higher education. The typology of the offerings is diverse, illustrating different learning scenarios.

The proposal from the University of Maryland, with a duration of one academic year, is aimed at first-year students in the first cycle. It is an extra-curricular programme, integrated into an optional research initiation scheme intended to provide an alternative learning experience.

The offering from the Università Ca' Foscari Venezia takes the form of a second-cycle specialisation programme, designated as a first-level Master's degree. It has a duration of one academic year and corresponds to 60 **ECTS**. The programme is structured into eleven curricular units, covering topics in Quantum Computing, Machine Learning (classical and quantum) and Applications. No information is publicly available regarding the detailed syllabi of these units.

The proposals from University College Dublin and the University of Oslo correspond to curricular units within second-cycle programmes, offered by the Departments of Computer Science and Physics, respectively.

The Halmstad University proposes an optional third-cycle curricular unit.

The publicly available information on each of these curricular units is limited. A careful analysis of the respective websites made it possible to identify a set of competences, syllabus contents and recommended bibliography, presented in Tables 4, 5 and 6, respectively. The items marked in these tables correspond only to those for which explicit information was found. For example, it is difficult to conceive that data representation is not addressed in the context of **QML**; however, if no explicit public information is available regarding this topic, it is not marked in the tables.

The information presented here is used in Chapter 4 to contextualise the concrete proposal of the curricular unit that is the object of this report.

It is possible to find a wide range of **Quantum Machine Learning** courses on *online* learning platforms. These courses are typically offered by a higher education institution in partnership with companies. Table 7 provides an overview of some of these courses.

***Online* Courses**	
Quantum Machine Learning (EDX/University of Toronto)	Data representation, variational circuits, quantum *kernels* and *feature maps*, quantum generative algorithms, quantum Gaussian processes.
Quantum Machine Learning (OpenHPI)	Quantum *Support Vector Machines*, quantum variational classifiers, quantum generative algorithms.
Quantum Computing and Introduction to Quantum Machine Learning (Udemy)	quantum *clustering*, quantum *Support Vector Machines*.
Quantum Machine Learning (Blue Courses)	Data representation, quantum *Support Vector Machines*, adiabatic quantum computing, quantum variational algorithms, quantum combinatorial optimisation.

Table 7: *Online* Courses

In 2021, IBM organised the 2021 Qiskit Global Summer School on Quantum Machine Learning. This course is a key reference, given the volume of supporting material provided and the breadth of the topics covered. In addition to the lectures on the various subjects, it also includes, in the form of *Jupyter notebooks*, the implementation, training and use of the different algorithms using **Qiskit**. The main topics addressed include:

- data representation;
- variational circuits;
- *barren plateaus* and limits on the trainability of quantum variational models;
- quantum combinatorial optimisation;
- quantum *kernels* and *feature maps*;
- quantum *Support Vector Machines*;
- quantum generative algorithms.

Following the tradition of Summer Schools, we have decided to include in this report a reference to the Quantum Machine Learning School, organised by the Ukrainian Catholic University, with instructors drawn from several institutions and companies around the world.

2.3 National Context

Over recent years, educational provision in the field of **Quantum Computing** has also increased in Portugal across the various cycles of higher education. In some cases, this provision is quite broad, targeting Quantum Technologies, which include Quantum Computing, Information Physics, Quantum Communications, the Quantum Internet, and Quantum Sensing and Metrology. Specific provision in the field of **QML** is still non-existent, with the exception of the curricular unit that is the object of this report. Table 8 summarises the educational proposals identified in the domain of **Quantum Computing**.

Higher Education in Quantum Computing in Portugal

University	Designation	Cycle	ECTS	Observations
Beira Interior	Physics and Information	1^{st}	6	Informatics Engineering
Aveiro	Quantum Computing	2^{nd}	24	Specialization
Coimbra	Quantum Computing	2^{nd}	6	Electrotechnique and Computers Engineering (optional)
Lisboa (IST)	Quantum Sciences and Technologies	2^{nd}	18	Minor for Masters in Sciences or Engineering
Porto	Introduction to Quantum Information	2^{nd}	6	Master in Physics
Porto	Quantum Computing	3^{rd}	5	Doctoral Programme in Informatics
Coimbra	Quantum Computing and Technologies	–	12	Specialization – not an academic degree

Table 8: Higher Education in Quantum Computing – Portuguese landscape

It is possible to identify, across the various proposals listed in Table 8, a common core of syllabus contents related to Information Physics, fundamental concepts of Quantum Computing and quantum algorithms. The fundamental concepts include the notion of the *qubit*, superposition, entanglement, the no-cloning theorem, quantum *gates* and circuits. The subset of algorithms addressed is usually selected from Simon's algorithm, Deutsch–Jozsa, the quantum Fourier transform, Shor, Grover and/or quantum amplitude estimation. These syllabi contents are therefore closely aligned with the recommendations of the **ACM**

Computer Science Curricula 2024 [3] (see Section 2.1, Table 1). The proposal from the University of Aveiro stands out for also addressing logics for new programming paradigms, namely variants of Dynamic Logic for the verification of hybrid and quantum programmes. This content is also covered by the educational provision of the University of Minho, within the specialisation in Information Physics of the Master's Degree in Physical Engineering.

2.4 Synthesis

In recent years, there has been a significant increase in educational provision in the domain of **Quantum Computing** across all cycles of higher education, both nationally and internationally. Education in this field is also supported by the most recent recommendations of the **ACM** curricula.

The situation is markedly different with regard to **Quantum Machine Learning**. Educational provision in higher education is scarce, and in Portugal it is non-existent apart from the curricular unit that is the object of this report.

Chapter 3

Curricular Context

This chapter presents the curricular context of the **Curricular Unit (CU) Quantum Machine Learning** within the Master's degree in Physical Engineering, specialisation in Information Physics. In particular, this **CU** is positioned within the education and training in **Quantum Computing** offered to students of this programme.

This educational provision is characterised within the framework of the **European Competence Framework in Quantum Technologies** (**ECFQT**) [1], developed by the Quantum Technology Education project, within the context of the Quantum Flagship initiative promoted by the European Union. Within the same framework, the graduate profile for this area of specialisation is also characterised.

3.1 European Competence Framework for Quantum Technologies

This framework, referred to by its English acronym **European Competence Framework in Quantum Technologies (ECFQT)**, proposes a mapping of possible competences and knowledge in **Quantum Technology (QT)**, as well as a characterisation of graduate and professional profiles in the field.

The **ECFQT** constitutes a taxonomy of content and proficiency levels in Quantum Technologies, facilitating the design and comparison of educational projects and competence profiles. This reference framework is organised into three dimensions: content map, proficiency triangle, and qualification profile. These dimensions are presented in the following subsections.

The **ECFQT** was developed and is updated and maintained by the Quantum Technology Education project, within the context of the Quantum Flagship initiative promoted by the European Union.

The European Quantum Readiness Center, an initiative of the Quantum Flagship, aims to establish a set of good practices in the areas of education and research in Quantum Technologies, promoting the **ECFQT** as a standardisation and communication tool.

The latest version dates from April 2024 [1, 5, 6]. The documents and tools made available within the scope of the **ECFQT** are covered by a CC BY 4.0 licence and are used in this report in accordance with that licensing.

3.1.1 Content map

The content map provides a structured view of content relevant to Quantum Technologies.

It is organised into eight domains with 42 subdomains. These subdomains are further detailed into topics and subtopics. Figure 1 presents the domains and subdomains. The complete map, including topics and subtopics, can be consulted in [1].

The domains are grouped into three subsets, represented using different colours, corresponding to the three areas of proficiency identified in the following subsection.

3.1.2 Proficiency

Proficiency is structured along two dimensions: area and level.

The **ECFQT** specifies three areas of proficiency encompassing different competences in **Quantum Technology**. The proficiency area does not depend directly on any particular set of quantum technologies; rather, it depends on the type of competences in relation to any quantum technology.

The areas of proficiency are:

I – Quantum concepts – encompasses knowledge and competences in quantum mechanics, including mathematical and physical formalisms. It corresponds to the green region in Figure 1, i.e. domains 1 and 2;

II – Hardware and software engineering – competences and knowledge related to the use, design and development of quantum technologies. It includes practical tasks (such as using a quantum computer) as well as conceptually more demanding tasks, such as the development and integration of quantum technologies. It corresponds to the blue region in Figure 1, i.e. topics from domains 3, 4, 5, 6 and 7;

III – Applications and strategies – covers the valorisation of quantum technologies (domain 8 of the content map) and addresses the business dimension of their applications (subdomains 5.6, 6.7, 7.4, 7.6). It includes competences related to the market, use cases, strategy development, ethics, responsibility and education. It corresponds to the bordeaux region in Figure 1.

Figure 1: **ECFQT** – content map: domains and subdomains

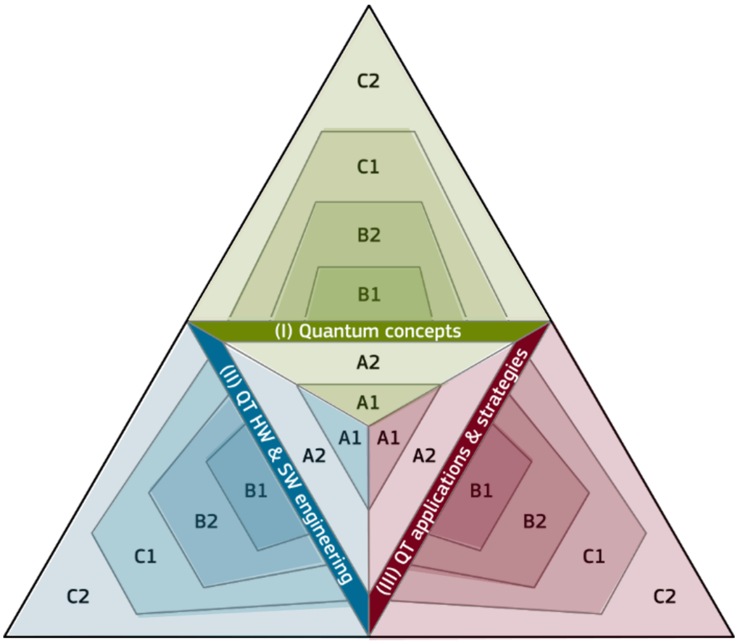


Figure 2: **ECFQT** - proficiency triangle (reproduced from [1])

Inspired by the structure adopted by the Common European Framework of Reference for Languages, the **ECFQT** reference framework proposes six proficiency levels: **A1 – Awareness**, **A2 – Literacy**, **B1 – Usage**, **B2 – Research**, **C1 – Specialisation**, and **C2 – Innovation**.

The proficiency triangle (Figure 2) relates proficiency areas and levels, enabling an integrated visualisation of the profile of an educational project or a Quantum Technologies professional. Its purpose is to convey a first impression of the qualifications associated with an individual, required for a job role or developed by a course, independently of the specific **Quantum Technology**.

The training begins at the centre, level A1, and progresses to level A2, covering the fundamental competences. This level of proficiency, associated with the central inverted triangle in Figure 2, can be achieved with a few days of training or self-study.

Proficiency at levels B1 and B2 is associated with the ability to use and conduct supervised research in one or more quantum technologies. It requires months of focused effort and is typically associated with a bachelor's degree (B1) or a master's degree without a dissertation in the area (B2).

Level C proficiency corresponds to specialisation in the field, requiring years of study and independent research. Level C1 is associated with a master's degree with a dissertation in the area, and level C2 with the completion of a doctoral degree. Proficiency levels may also be achieved through professional learning, supported by mentors and underpinned by research and experience.

Table 9 reproduces the characterisation of the various proficiency levels for each area, according to [1].

I – Quantum concepts	
A1 – Awareness	**Reproduce** basic concepts and terminology
A2 – Literacy	**Describe** fundamental concepts
B1 – Usage	**Apply** quantum methods to problems
B2 – Research	**Analyse** problems using quantum methods
C1 – Specialisation	**Refine** quantum methods
C2 – Innovation	**Develop** innovative solutions

II – Hardware and software engineering	
A1 – Awareness	**Reproduce** basic functionalities of one or more **QT**
A2 – Literacy	**Perform** basic tasks using **QT**
B1 – Usage	**Apply** **QT**
B2 – Research	**Analyse** performance and **optimise** the use of a **QT**
C1 – Specialisation	**Design** the integration of **QT**
C2 – Innovation	**Develop** new **QT**

III – Applications and strategies	
A1 – Awareness	**Recognise** the potential of **QT**
A2 – Literacy	**Identify** the value associated with **QT**
B1 – Usage	**Classify** approaches and applications of **QT**
B2 – Research	**Analyse** the market and opportunities for **QT**
C1 – Specialisation	**Advise** on the selection and application of **QT**
C2 – Innovation	**Develop** and **evaluate** market strategies related to **QT**

Table 9: **ECFQT**: proficiency levels in **Quantum Technology (QT)** (reproduced from [1])

3.1.3 Qualification profiles

The **ECFQT** proposes nine example qualification profiles in Quantum Technologies, corresponding to typical profiles and areas of proficiency. Their respective proficiency triangles allow for rapid comparison of different profiles—see Figure 3.

The profiles must be combined with a selection of topics from the content map, resulting in an association with a set of Quantum Technologies.

These profiles will be used to characterise graduates of the **MEPhys**, specialisation in **Physics of Information**, with and without a dissertation in the area of **Quantum Computing**.

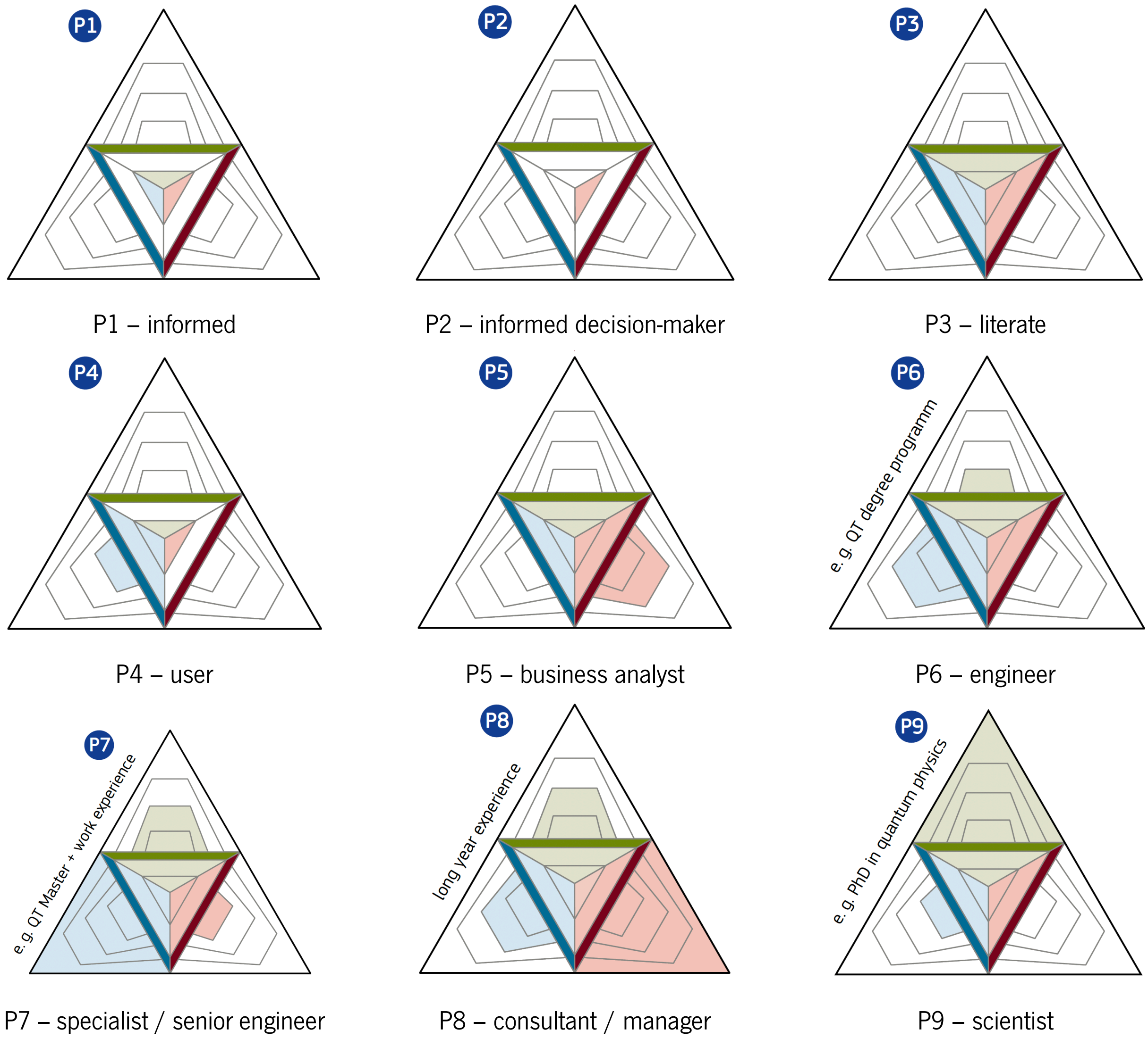


Figure 3: **ECFQT**: qualification profiles

3.2 Education in Physical Engineering

The relevance of Physics to the global economy in general, and to the European economy in particular, has steadily increased over recent decades. With the aim of clarifying and quantifying this relevance, the European Physical Society sponsored a study in 2019 to address questions such as: i) what is the turnover of the European industrial sector linked to Physics; ii) what percentage of total European employment is in this same sector.

The final report of this study [7], based on data from 31 European countries between 2011 and 2016, presents conclusive results. The physics-based industrial sector generated 16% of the total turnover considered during this period, amounting to approximately 4 billion euros ($\sim 4 \times 10^{12}$ €). It is also responsible for 12% of employment in Europe. This sector is highly research- and development-intensive, requiring investment in research and in the training of highly qualified human resources.

Aware of the relevance of this field of knowledge, the University of Minho offers degrees at two cycles of higher education: the **BsC in Physical Engineering (LEPhys)**, corresponding to the first cycle, and the **Master in Physical Engineering (MEPhys)**, corresponding to the second cycle. The latter provides the technical and scientific capacity required for access to third-cycle programmes (Doctorate).

3.2.1 Bachelor's Degree in Physical Engineering

The **LEPhys** degree provides students with a solid foundation in physics, mathematics, electrical engineering, and computer science, primarily preparing them to continue their studies in the **MEPhys**. It also equips them with knowledge compatible with a broad range of second-cycle engineering programmes, requiring only minimal complementary training.

Table 10 presents the curriculum of this programme. The Course Units most relevant to the area of **Quantum Computing** are shaded in violet.

Bachelor's Degree in Physical Engineering

Year 1					
Semester 1			**Semester 2**		
Linear Algebra and Analytical Geometry for Sciences	M	6	Complements of Calculus and Analytical Geometry	M	6
Calculus for Sciences	M	6	Vector Calculus	M	6
Introduction to Experimental Physics	Phys	6	Circuit Analysis	EEIC	6
Introduction to Functional Programming	I	6	Data and Computation	I	6
General Chemistry	Chem	6	Newtonian Mechanics	Phys	6
Year 2					
Semester 3			**Semester 4**		
Complex and Fourier Analysis	M	6	Statistical Physics	Phys	5
Introduction to Electronics	EEIC	6	Electromagnetism Laboratories	Phys	5
Electromagnetism	Phys	6	Complements of Electronics and Digital Systems	EEIC	5
Newtonian Mechanics Laboratory	Phys	6	Quantum Mechanics	Phys	5
Analytical Mechanics and Waves	Phys	6	Imperative Programming	I	5
–			Computer Systems	I	5
Year 3					
Semester 5			**Semester 6**		
Complements of Quantum Mechanics	Phys	5	Solid State Physics	Phys	5
Computational Physics for Engineers	Phys	5	Instrumentation	EEIC	5
Introduction to Silicon Microtechnologies	EEIC	5	Introduction to Optics	Phys	5
Parallel Programming	I	5	Integrated Sensors	EEIC	5
Option I			Option II		
• Algorithms and Complexity	I	5	• Quantum Information	Phys	5
• Advanced Topics in Electromagnetism	Phys	5	• Control Theory	EEIC	
Continuous and Discrete Signal Processing	EEIC	5	UMinho Option	–	5

Table 10: **BsC in Physical Engineering** – study plan

The syllabus of the Course Units relevant to Quantum Computing includes:

Quantum Mechanics – Dirac notation, quantum entanglement, eigenvalues, the concept of the Hamiltonian, and the time evolution of states.

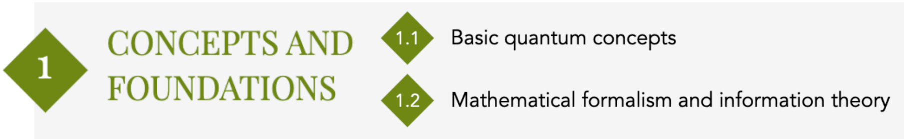


Figure 4: Quantum Mechanics – Domains and sub-domains

Complements of Quantum Mechanics – many-body quantum physics.

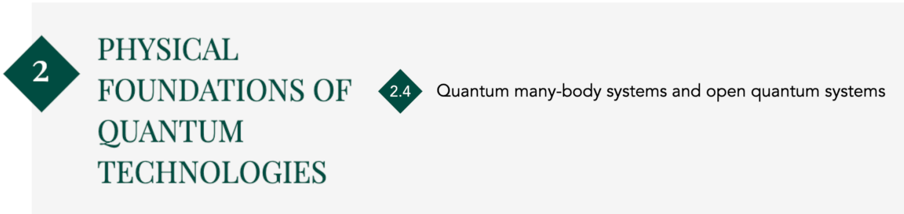


Figure 5: Complements of Quantum Mechanics – Domains and sub-domains

Quantum Information – classical information theory, foundations of quantum theory, qubits and their physical implementation, quantum entanglement, quantum information and its processing.

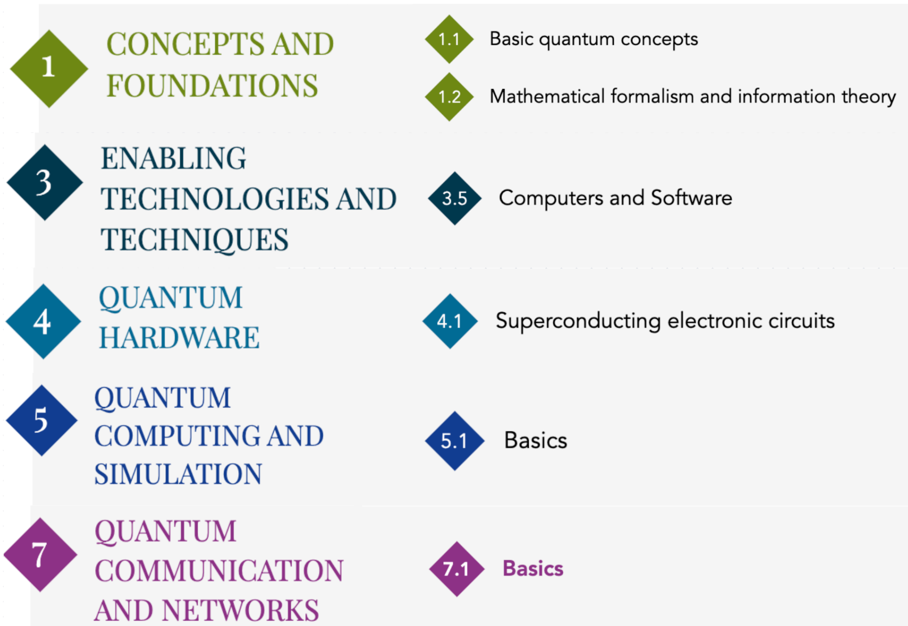


Figure 6: Quantum Information – Domains and sub-domains

This training is the responsibility of the Department of Physics and provides a solid foundational education for students interested in pursuing further studies in Information Physics and Quantum Computing.
A graduate in Physical Engineering from the University of Minho exhibits the qualification profile in Quantum Computing presented in Figure 7.

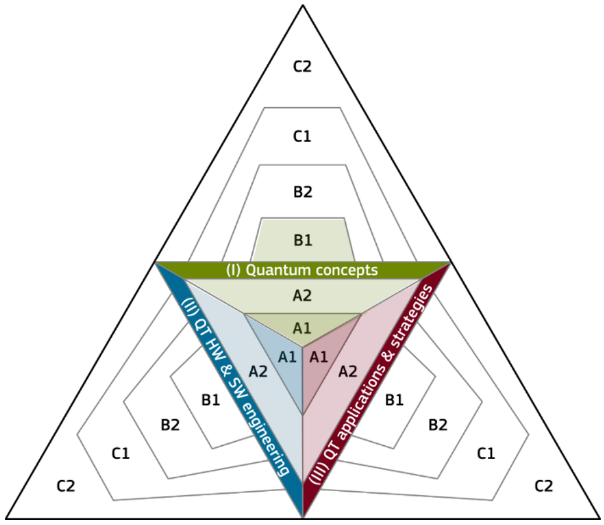


Figure 7: Qualification profile of the Physical Engineering graduate from the University of Minho

This graduate training addressed the domains highlighted in Figure 8, and his proficiency levels in **Quantum Computing (QC)** are summarised in Table 11.

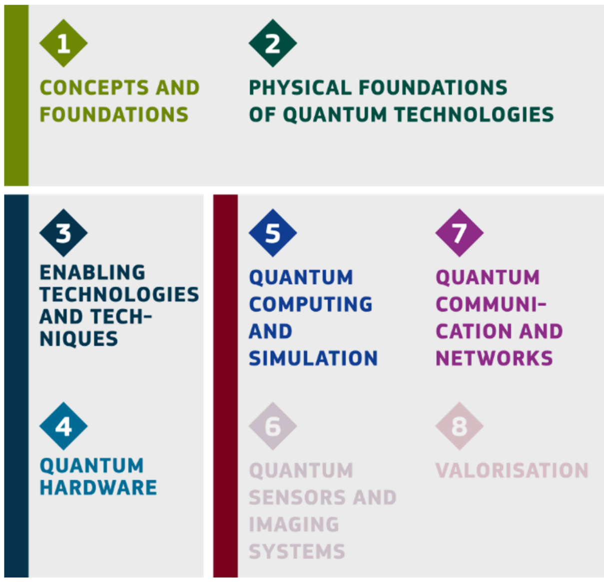


Figure 8: Quantum Technology content domains of the Physical Engineering graduate from the University of Minho

3.2.2 Master's Degree in Physical Engineering

The Master's Degree in Physical Engineering at the University of Minho offers two specialisation tracks: **Devices, Microsystems and Nanotechnologies (DMN)** and **Physics of Information (PI)**, sharing a common core of 6 **CU** (out of 15), corresponding to 75 ECTS (out of 120).

I – Quantum concepts	
B1 – Usage	**Apply** quantum methods to problems

II – Hardware and software engineering	
A2 – Literacy	**Perform** basic tasks using **QC**

III – Applications and strategies	
A2 – Literacy	**Identify** the value associated with **QC**

Table 11: Quantum Computing proficiency levels of the Physical Engineering graduate from the University of Minho

The latter track concentrates the entire educational offer in the area of **Quantum Computing**. The **CU** addressed in this report integrates the study plan of the **Physics of Information** specialisation; therefore, only this track is discussed throughout this document.

The **PI** track includes the following learning outcomes:

- Use quantum physics as a computational paradigm to efficiently compute programs otherwise not accessible;
- Develop quantum algorithms in dedicated programming languages;
- Develop multidisciplinary physical engineering projects exploring interfaces between scientific areas.

Table 12 presents the study plan of this programme. The Course Units most relevant to **Quantum Computing** are shaded in violet, with the **CU** presented in this report highlighted in green. In the current curriculum, this **CU** is entitled **Quantum Data Science**. As proposed in Chapter 1, this designation should change to **Quantum Machine Learning** in the near future. This latter designation is used throughout this report, except where the current curriculum is explicitly referenced.

The syllabi of the Course Units framed within the area of **Quantum Computing** include:

Topics in Quantum Information – this is an optional **CU**, whose objective is to introduce the fundamentals of Quantum Information to students who do not have prior training in the area, namely those who have not attended the **LEPhys** programme or have not selected the “Quantum Information” option in its third year.

Master's Degree in Physical Engineering: Information Physics

Year 1					
Semester 1			**Semester 2**		
Quantum Computing	I	5	Quantum Logic	I	5
Probabilistic Programming	I	5	Communications and Networks	I	5
Language Theory and Compilation	I	5	Cryptography and Information Security	I	5
Advanced Instrumentation	EEIC	5	Quantum Data Science	I	5
Option • Topics in Quantum Information • Topics on Physical Platforms for Quantum Computing	 Phys Phys	 5 5	Cyber-Physical Computing	I	5
Photonics I	Phys	5	Photonics II	Phys	5
Year 2					
Semester 3			**Semester 4**		
Business Training and Entrepreneurship	EIS	5			
Applied Physical Engineering	EFis	10			
Dissertation				EFIS	45

Table 12: **Master in Physical Engineering** – study plan: Information Physics

- Introduction to quantum mechanics as a generalisation to complex probability amplitudes, wave–particle duality, superposition and interference;
- Information and bits; examples of 1-qubit systems; photons, polarisers and interferometers; spin-½ particles; two-level atoms;
- Quantum states and observables; Hilbert space; operators; observables; the dual space; eigenvalues and eigenvectors;
- Distinguishability of quantum states; the Born projection rule; the uncertainty principle; Hermitian operator formalism for measurements;
- Quantum dynamics; the Schrödinger equation; unitary evolution; operators and symmetries; some quantum gates: Pauli, Hadamard, CNOT;
- Entanglement; composite systems; interaction as a source of entanglement; conditional states; the

EPR paradox; Bell's theorem; GHZ states;

- Information and e-bits; the no-cloning theorem; distinguishability and decoding; quantum key distribution;
- Density operators; statistical mixtures; the interior of the Bloch sphere; Bloch equations;
- Quantum entropy; quantum data compression.

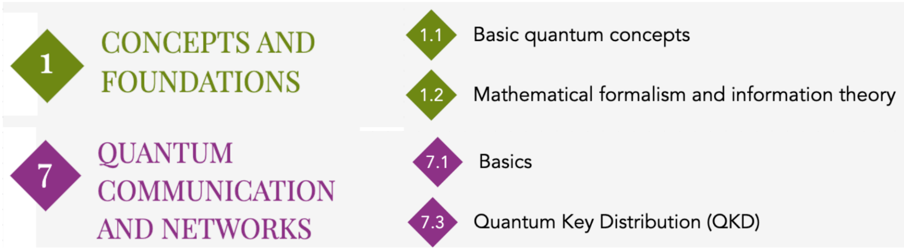


Figure 9: Topics in Quantum Information – Domains and sub-domains

Topics on Physical Platforms for Quantum Computing – this is an optional **CU**, aimed at students who already have some background, namely at first-cycle level, in Quantum Information.

- Historical development and results in the manipulation of single quantum systems;
- DiVincenzo criteria;
- Model Hamiltonians common to trapped ions, quantum optics, and interacting quantum dots;
- Qubit implementations, their dynamics, and initialisation/readout in different platforms;
- Quantum control techniques for error mitigation and decoherence control;
- State-of-the-art implementation of quantum algorithms and other protocols on different physical platforms, and scalability perspectives.

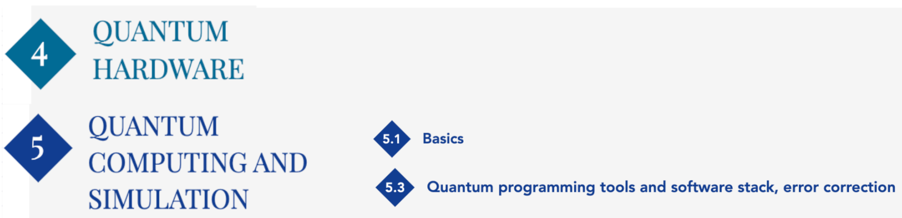


Figure 10: Topics on Physical Platforms for Quantum Computing – Domains and sub-domains

Quantum Computing

- Review of notions of computability, Turing machines, lambda calculus;
- Models for quantum computation: quantum Turing machines; quantum lambda calculus;

- Computational complexity of quantum programs;
- Quantum algorithms: basic principles; Deutsch–Jozsa, Grover and variants; Quantum Fourier Transform; phase estimation algorithms; the hidden subgroup problem; Shor's algorithm;
- Quantum simulation techniques;
- Quantum programming: Quipper and Qiskit languages.

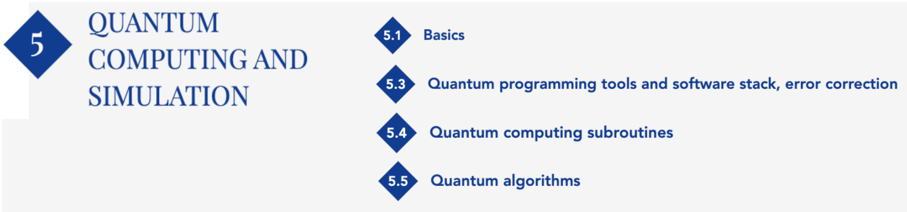


Figure 11: Quantum Computing – Domains and sub-domains

Quantum Logic

- Foundations: categorical semantic models, universal properties and adjunctions, monoidal categories; applications: (categories of) relations, matrices and Hilbert spaces;
- Computational logics and semantics for quantum computation; computational interpretation of the basic postulates of quantum mechanics;
- Corresponding categorical structures: monoidal (composition), compact closed (entanglement), adjunctions (inner product), biproducts (non-deterministic branching);
- Linear lambda calculus and the Curry–Howard–Lambek correspondence for quantum computation;
- Diagrammatic reasoning for the derivation and modelling of quantum algorithms;
- Modelling and verification of quantum algorithms.

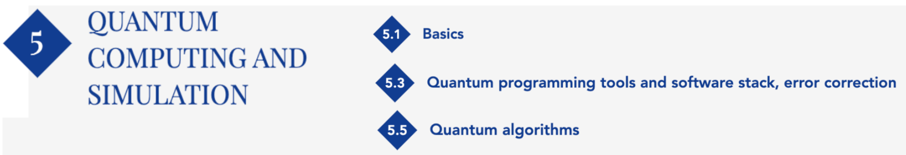


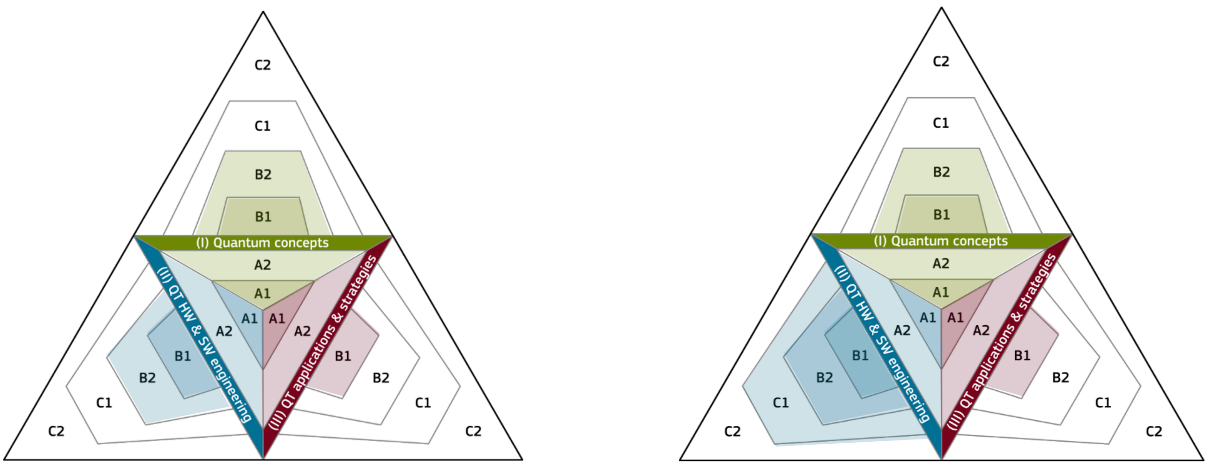
Figure 12: Quantum Logic – Domains and sub-domains

Quantum Data Science / Quantum Machine Learning – this **Curricular Unit** is the subject of this report. Its planning is detailed in Chapter 4. For completeness, a brief description of the syllabus is included in this section. The **ECFQT** makes no explicit reference to **Quantum Machine Learning**. It appears appropriate to place it within sub-domain **5.6 – Applications of quantum computing**, with the main subroutines and algorithms framed, respectively, within sub-domains **5.4 – Quantum subroutines** and **5.5 – Quantum algorithms**. The placement of this **CU** in the second semester of the first year ensures that students have a solid background in Quantum Mechanics, Quantum Information and Quantum Computing, which is essential for successfully achieving the proposed learning outcomes.

- Introduction to classical machine learning;
- Quantum data representation;
- Variational quantum classification; expressivity and trainability;
- Combinatorial optimisation: **QAOA** and **QUBO**;
- Quantum *feature maps* and *kernels*; hybrid **SVM** algorithm.

Figure 13: Quantum Data Science / Quantum Machine Learning – Domains and sub-domains

The qualification profile in Quantum Computing and the proficiency levels of a Master in Physical Engineering, specialising in Information Physics, depend on whether or not the dissertation is carried out in this area. These are presented in Figure 14 and Table 13.



(a) Without a dissertation in **QC**

(b) With a dissertation in **QC**

Figure 14: Qualification profile of the Master in Physical Engineering, specialising in Information Physics, at the University of Minho

This qualification profile corresponds to profiles P6 / P7 of the **ECFQT** (see Figure 3), aptly designated by this framework as the profile of a *software* engineer in **Quantum Computing**. A professional with this

Without dissertation		With dissertation	
I – Quantum concepts			
B2 – Research	**Analyse** problems with quantum methods	B2 – Research	**Analyse** problems with quantum methods
II – Hardware and software engineering			
B2 – Research	**Analyse** problems with quantum methods	C1 – Specialisation	**Design** the integration of **QC**
III – Applications and strategies			
B1 – Usage	**Classify** approaches and applications of **QC**	B1 – Usage	**Classify** approaches and applications of **QC**

Table 13: Quantum Computing proficiency levels of the Master in Physical Engineering at the University of Minho

profile has extensive knowledge of the concepts and mathematical formalisms associated with quantum computing. Their role typically consists of the integration of quantum algorithms into applications or *software* libraries.

3.3 Summary

The Master's Degree in Physical Engineering, specialising in Information Physics, offered by the University of Minho, trains software engineers in the field of Quantum Computing, with a solid grounding in the mathematical and physical foundations associated with this area of knowledge. The characterisation of the qualification profile of these graduates using the **ECFQT** reference framework clearly demonstrates its alignment with the recommended profile for an engineer in Quantum Technologies, both in terms of content areas and proficiency levels — compare Figures 3 (P6 and P7) and 14.

The training offered to students in the areas of **Quantum Computing** and **Physics of Information** ensures the relevant knowledge and skills to support the **Curricular Unit Quantum Data Science**, the subject of this report, offered in the second semester of the first year of the **MEPhys**. This is not the case with regard to knowledge in the area of Machine Learning, as detailed in Chapter 4.

The characterisation, in Chapter 2, of the educational offer at other institutions, together with the characterisation of the training in **MEPhys–PI** presented in this chapter, highlights the unique and innovative nature of the latter. Indeed, this is the only educational project (identified by the author) that covers, with some depth, six of the eight domains of the content map proposed by the **ECFQT**. The emphasis is placed on Domain 5 (Quantum computing and simulation), reinforcing its innovative and comprehensive character by including specific training in the areas of Quantum Logic and Quantum Machine Learning. The identification of domains (and sub-domains) not addressed, or addressed with less depth, may be used as an auxiliary tool to guide future restructuring of this educational offer. In particular, Domain 8 (Valorisation) stands out — this component relating to impact, market, and ethics could benefit from explicit treatment integrated into the course curriculum.

Chapter 4

Curricular Unit: Quantum Machine Learning

This chapter presents the description of the **Curricular Unit Quantum Machine Learning**, which is the subject of this report. Some results and reflections arising from the three editions delivered in the academic years 2021/22, 2022/23 and 2023/24 are also presented.

4.1 Description

This section characterises the organisation of this **CU**, placing it in context, presenting its objectives and describing the adopted teaching/learning process.

4.1.1 Context

This **Curricular Unit** is part of the first year / second semester of the study plan of the **Master in Physical Engineering**, within the specialisation area of **Physics of Information**, and was originally introduced under the designation **Quantum Data Science**. As discussed in Chapter 1, it is intended that, at the next opportunity, this designation will be changed to **Quantum Machine Learning**, and this will therefore be the formulation used throughout this chapter.

Students attending this specialisation who obtained their bachelor's degree by completing the **BsC in Physical Engineering** will already have taken (Table 10):

Quantum Mechanics and Complementary Topics in Quantum Mechanics – fundamental concepts of quantum mechanics; mathematical tools and abstractions required to operate in the Hilbert spaces inhabited by quantum states and operators;

Quantum Information – classical and quantum information theory; qubits and their physical implementation; quantum superposition, entanglement and interference.

Students enrolled in the **MEPhys** who have not attended the **LEPhys** must necessarily satisfy the following admission criteria for this programme:

a. holders of a Bachelor's degree in Physical Engineering, Physics, Electronic or Electrical Engineering, Computer Engineering or related areas;
b. holders of a foreign higher education degree awarded following a first cycle of studies organised in accordance with the Bologna Process by a State adhering to this process, in the areas of knowledge referred to in a);
c. holders of a foreign higher education degree that is recognised by the Scientific Council of the School of Engineering as satisfying the objectives of a bachelor's degree, in the areas of knowledge referred to in a);
d. holders of an academic, scientific or professional curriculum that is recognised by the Scientific Council of the School of Engineering as attesting the capacity to undertake this cycle of studies.

Given these criteria, it is expected that these students will have some background in Quantum Mechanics, enabling them to successfully attend this second-cycle programme. In the first semester of the **MEPhys-PI**, an optional **CU** is offered whose purpose is precisely to provide these students with a solid foundational education in Information Physics:

Topics in Quantum Information – concepts of quantum mechanics; quantum entanglement, superposition and interference; information and qubits; Hilbert space, operators and observables.

During the first semester of the first year, students of the **MEPhys** receive training in:

Topics on Physical Platforms for Quantum Computing – requirements of a physical platform for quantum information processing (DiVincenzo criteria); model Hamiltonians; qubit implementations;

Quantum Computing – notions of computability; models for quantum computation; computational complexity of quantum programmes; quantum algorithms: Deutsch–Jozsa, Grover, Fourier transform and Shor's algorithm; quantum programming: Quipper and Qiskit languages.

At the beginning of the second semester of the first year, students have therefore been exposed to training that ensures, in the areas of **Quantum Computing** and **Physics of Information**, the relevant knowledge and skills required to support the **Curricular Unit Quantum Machine Learning**.
This is not the case, however, with regard to knowledge in the area of **Machine Learning (ML)**. In fact, no **CU** offered within the context of the **LEPhys** or **MEPhys** addresses the fundamental competences and contents associated with **ML**. This deficit in students' prior training requires the explicit inclusion of these contents in the lesson plan of this **CU**, which consequently constrains its objectives, learning outcomes and syllabus (see subsections 4.1.2 and 4.1.4).

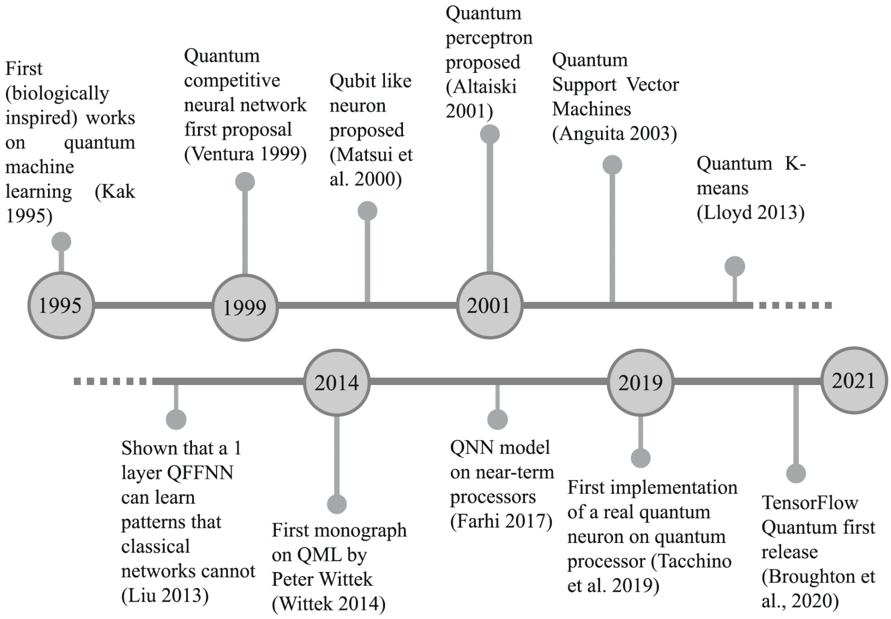


Figure 15: Timeline of the development of **QML**. Published in [2].

4.1.2 Objectives and learning outcomes

Quantum Machine Learning is a very recent area of knowledge, as illustrated in Figure 15. The designation **Quantum Machine Learning** itself first appeared in 2013 [8], and the first book devoted to this field was published in 2014 [9]. It is therefore a highly dynamic domain, with the continuous publication of new results and revisions of previous ones. There is not yet a set of established practices, nor practical demonstrations of any advantage of **Quantum Machine Learning** over classical **Machine Learning**.

The objective of this **CU** therefore cannot be the development of competences for the efficient solution of problems using **QML**—it has yet to be determined whether such efficiency is achievable and to establish a set of practices that would enable its potential realisation.

The main objectives are:

O1 – to convey knowledge and develop competences to design and evaluate solutions in the domain of **Quantum Machine Learning**;

O2 – to promote critical questioning of results obtained using **QML**, identifying advantages/disadvantages (relative to classical **ML**) and limits inherent to quantum computation and its technological realisations.

A student who successfully completes this curricular unit should be able to:

LO1 – identify, within the area of **Machine Learning**, opportunities for the use of quantum resources;

LO2 – design quantum computing solutions for problems related to data exploration;

LO3 – implement hybrid (classical + quantum) learning algorithms;

LO4 – identify limitations (current and medium-term) associated with **QML**;

LO5 – demonstrate written and oral communication skills appropriate for conveying approaches, expectations, results and limitations, promoting language as a tool for the expression and construction of thoughts, ideas and arguments, grounded in critical thinking.

4.1.3 Teaching/learning process

The operation of the **CU** is organised over 15 weeks, with one lecture (commonly referred to as theoretical) and one tutorial (theoretical–practical) session per week, each lasting 2 hours. The associated workload corresponds to 5 ECTS credits, comprising 60 contact hours (lectures plus tutorials) and 80 hours of independent work (study and project development).

The lectures are essentially expository, supported by a set of slides (see subsection 4.1.6). Interaction with students is valued and actively promoted. The number of students admitted each academic year to the **MEPhys-PI** is limited to 20. The small class size facilitates interaction, with the lecturer frequently engaging students directly.

In addition to direct questioning, lectures are supported by the use of an **Audience Response System (ARS)**. The author attended the Docência+ course in September 2021 and the Avaliação+ course in October 2022, joint initiatives of the Universities of Aveiro and Minho aimed at staff development. These courses discussed various digital tools supporting different teaching/learning paradigms, leading the author to adopt an **ARS** as a catalyst for interaction among students and with the lecturer.

VoxVote is an **Audience Response System** that allows multiple-choice questions to be posed to a large audience. Each student selects their answer on their local device; all responses are collected by the server and subsequently displayed for discussion. The use of such systems during lectures encourages student engagement and contributes very significantly to active participation.

This system was used in several lectures, typically 15 to 20 minutes before the end of the session. In each of these sessions, two to four multiple-choice questions were presented, some with multiple correct answers. The latter fostered direct discussion among students, contributing significantly to their engagement in the classroom dynamics.

Figure 16 presents an example of a multiple-choice question used in the data representation module. Appendix A presents the full set of questions prepared for this **CU**.

The tutorial sessions have a tutorial character, involving guided small to medium-sized problem solving,

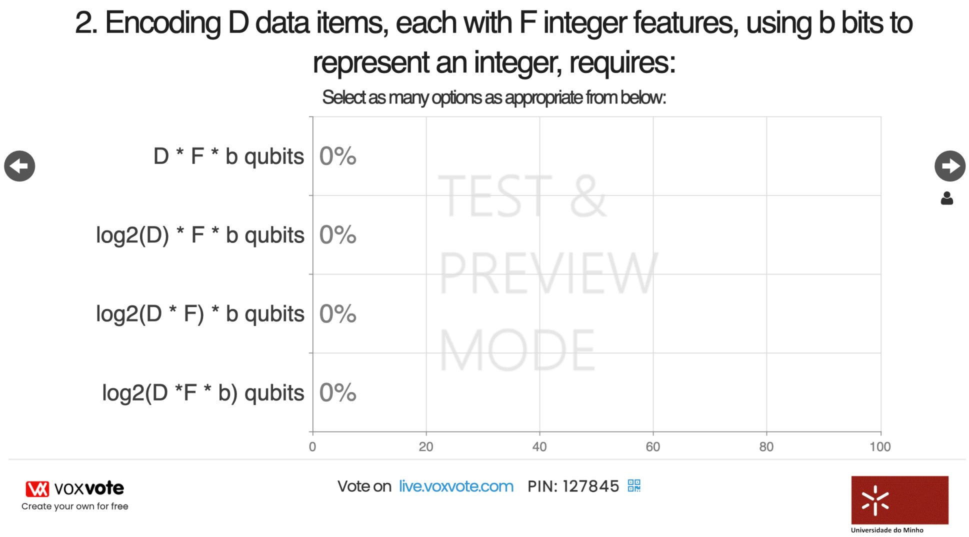


Figure 16: Example of VoxVote usage

using well-defined algorithms. These tutorials also require the evaluation of the adopted solutions and comparison with alternatives. Students use tools that enable the specification and execution/simulation of quantum programmes, as well as the visualisation of processed datasets. All tutorials were made available as Jupyter notebooks, enabling the presentation of interactive materials as well as the coding and execution of problems and their respective solutions (see subsection 4.1.6). Qiskit is used, a development framework for quantum systems based on the circuit model, which allows local execution using simulators and access to remote quantum machines via the IBM Quantum Platform.

4.1.4 Syllabus

The syllabus proposed for this **CU** aims to ensure that students successfully achieve the previously established learning outcomes — see Section 4.1.2. Consequently, the topics addressed cover data representation, the most representative quantum (and/or hybrid) algorithms, and their respective advantages and limitations.

As established in the previous chapters, students begin this **CU** with a deficit in training in classical Machine Learning. The fundamental concepts and algorithms associated with this field of knowledge are indispensable for the development of competences in **Quantum Machine Learning**. These are therefore introduced throughout the first four weeks of theoretical classes of this **CU**. As a result, the specific **QML** topics addressed are necessarily reduced in order to accommodate these initial four weeks. In particular, *Reinforcement Learning*, the *Ising* model, quantum vector operations (**quantum Basic Linear Algebra Subprograms (qBLAS)**), and algorithms commonly referred to as fault-tolerant are not covered.

Programme

1. **Classical Machine Learning: fundamentals**

   (a) Fundamental concepts: parameterised model, *loss*, (empirical) risk, generalisation;

   (b) optimisation, gradient method, convexity, saddle points and local minima;

   (c) linear and polynomial regression, *over (under) fitting*, regularisation;

   (d) classification, *perceptron* and logistic classifier, activation functions, *softmax*;

   (e) neural networks, universal approximators, *backpropagation*, *vanishing gradients*;

   (f) *Feature maps* and *kernels*

      i. the *kernel trick*, polynomial and Gaussian *kernels*, Gram matrix;

      ii. *Support Vector Machines*: primal and dual formulations;

      iii. *clustering*: *k-means*.

2. **Data representation**

   (a) *Basis encoding*;

   (b) *Angle encoding*;

   (c) *Amplitude encoding*;

   (d) *Higher-order encoding*.

3. **Variational quantum classification**

   (a) The variational principle, parameterised/variational quantum circuits (VQC);

   (b) variational classifiers, *ansätze*, observables, cost functions;

   (c) expressivity, *data reuploading*;

   (d) trainability, gradient estimation, *barren plateaus*.

4. **Combinatorial optimisation**

   (a) combinatorial optimisation;

   (b) adiabatic quantum computing, *Trotterisation*;

   (c) *Quantum Approximate Optimization Algorithm* — QAOA;

   (d) Quadratic programming and *Quadratic Unconstrained Binary Optimization* — QUBO.

5. **Quantum *kernels* and *feature maps***

(a) *data encoding* as a *feature map*;

(b) *quantum kernel estimation* (QKE), solutions accessible to VQCs as a subset of the solutions accessible to QKE;

(c) quantum *kernels* and quantum advantage.

4.1.5 Schedule

The plan for each of the weeks comprising the semester is presented below. Each week includes one theoretical lecture and one theoretical–practical (TP) session, both with a duration of two hours.

Week 1	
Lecture	**Quantum machine learning: discussion** (presentation) 1. **Classical machine learning: basics** (presentation) (a) basic concepts; (b) optimization, gradient method, convexity, saddle points, and local minima ; (c) linear and polynomial regression, *over (under) fitting*, regularization;
TP	***Swap test*** (jupyter notebook) – quantum computing basics, *Qiskit* programming. – Introduction to **QML**: using the swap test to develop a fundamental discriminatory algorithm.

Week 2	
Lecture	1. **Classical machine learning: fundaments** (presentation) (d) classification, perceptron, and logistic classifier.
TP	**Observables and expected value** (jupyter notebook) quantum computing basics, *Qiskit* programming.

Week 3	
Lecture	1. **Classical machine learning: fundaments** (presentation) (e) neural networks, back propagation, vanishing gradients;
TP	**Distance among states and entanglement** (jupyter notebook) quantum computing basics, *Qiskit* programming.

Week 4	
Lecture	1. **Classical machine learning: fundaments** (presentation) (f) Feature maps, kernels, Support Vector Machines.
TP	**Classical machine learning: perceptrons** (jupyter notebook)

Week 5	
Lecture	**Mini-test 1** 2. **Data representation** (presentation)
TP	**Classical machine learning: neural networks** (jupyter notebook)

Week 6	
Lecture	2. **Data representation** (presentation)
TP	**Classical machine learning: SVM** (jupyter notebook)

Week 7	
Lecture	3. **Variational Quantum Classifiers** (presentation) (a) The variational principle, parameterizable quantum circuits (PQC); (b) variational classifiers, cross entropy, *ansatze*;
TP	**Data representation** (jupyter notebook)

Week 8	
Lecture	3. **Variational Quantum Classifiers** (c) expressivity, data reuploading – (presentation); (d) trainability, barren plateaus – (presentation).
TP	**Variational Quantum Classifiers** (jupyter notebook)

Week 9	
Lecture	4. **Combinatorial Optimization** (presentation) (a) combinatorial optimization; (b) adiabatic quantum computing, trotterization; (c) Quantum Approximate Optimization Algorithm – QAOA;
TP	**VQC and barren plateaus** (jupyter notebook)

Week 10	
Lecture	**Mini-test 2** 4. **Combinatorial Optimization** (presentation) (d) Quadratic programming and Quadratic Unconstrained Binary Optimization – QUBO;
TP	**Project – support**

Week 11	
Lecture	5. **Quantum Kernels and Feature Maps** (presentation) (a) data encoding as a feature map; (b) quantum kernel estimation (QKE)
TP	**Quantum Approximate Optimization Algorithm – QAOA** (jupyter notebook)

Week 12	
Lecture	5. **Quantum Kernels and Feature Maps** (presentation) (b) quantum kernel estimation (QKE) (c) quantum kernels and quantum advantage.
TP	**Quantum kernels** (jupyter notebook)

Week 13	
Lecture	**Mini-test 3**
TP	**Project – support**

Week 14
Project – Final Presentation

4.1.6 Bibliography and supporting material

The main bibliographic reference is the book:

- Maria Schuld & Francesco Petruccione; "Machine Learning with Quantum Computers"; 2nd Edition; Series "Quantum Science and Technology"; Springer; ISBN 978-3-030-83097-7, December 2021

The authors are well-established researchers. Maria Schuld is one of the most recognised researchers in the field of **Quantum Machine Learning**.

The bibliographic reference supporting classical **ML** is the book:

- Luís Serrano; "Grokking Machine Learning"; Manning Publishers; ISBN 9781617295911, November 2021

The choice of a bibliographic reference that adopts an informal approach, less focused on the mathematical foundations underlying **ML** algorithms, is justified by the fact that this **Curricular Unit** does not include the development of competences in this area among its objectives. The intention is that students can understand fundamental concepts as support for the quantum approaches.

Supporting material

The teaching materials supporting the theoretical and theoretical–practical classes are available here and are also referenced in Section 4.1.5.

In addition, students are recommended some supplementary materials for a deeper approach to certain topics. These materials are particularly useful when they align with the theme of the project being assessed. The slide decks supporting each theoretical lecture include references to these materials, which are repeated here.

0 – Introduction

- "Is Quantum Advantage the Right Goal for Quantum Machine Learning?"; Schuld, Maria and Killoran, Nathan; PRX Quantum, vol 3(3), 2022

1 – Classical machine learning

- "Support Vector Machines: All you need to know!"
- "The Kernel Trick in Support Vector Machine (SVM)"
- "From Variational Classifiers to Linear Classifiers"; Qiskit Global Summer School 2021: Lectures 6.1, 2021
- Course Notes CS229: chapters 5 and 6; Andrew Ng

3 – Variational quantum classification

- "Building a Quantum Classifier"; Amira Abbas; Qiskit Global Summer School on Quantum Machine Learning: Lecture 5.1, 2021
- "Barren Plateaus, Trainability Issues and How to Avoid Them"; Tacchino; 2021 Qiskit Global Summer School on Quantum Machine Learning: Lecture 8.2, 2021
- "Training parameterized quantum circuits"; Qiskit Quantum Machine Learning Course

- ”A Review of Barren Plateaus in Variational Quantum Computing“; Larocca et al; May, 2024
- “The Effect of data encoding on the expressive power of variational quantum machine learning models”;Maria Schuld, Ryan Sweke and Johannes Jakob Meyer;Physical Reviews A, 2021
- “Input Redundancy For Parameterized Quantum Circuits”; Francisco Javier Gil Vidal, Dirk Oliver Jim Theis; 2020
- ”Reupload Qiskit Notebook”

4 – Combinatorial optimization

- ”Solving combinatorial optimization problems using QAOA”; Qiskit Textbook
- ”Theory and Implementation of the Quantum Approximate Optimization Algorithm: A Comprehensible Introduction and Case Study Using Qiskit and IBM Quantum Computers”
- ”Quantum Approximate Optimization Algorithm explained”
- ”Quantum TSP tutorial”
- ”Quantum Bridge Analytics I: A Tutorial on Formulating and Using QUBO Models”; Fred Glover, Gary Kochenberger, Yu Du

5 – Quantum kernels and feature maps

- “Supervised learning with quantum enhanced feature spaces”; Vojtech Havlicek et al.; 2018
- Qiskit Global Summer School 2021: Lectures 6.1, 6.2, 7.1, 10.2; 2021
- “A rigorous and robust quantum speed-up in supervised machine learning”; Yunchao Liu, Srinivasan Arunachalam, Kristan Temme; Nature Physics volume 17, 2020

4.1.7 Assessment methodology

Assessment is carried out using two distinct instruments: written mini-tests and the completion of a project. The written mini-tests contribute 40% of the student’s final mark. Three are held throughout the semester to assess students’ assimilation of the concepts conveyed in the lectures. The first mini-test covers topic **1** of the syllabus (see subsection 4.1.4); the second covers topics **2** and **3**; the third covers topics **4** and **5**. Appendix B includes the three model mini-tests that were made available to students in the 2023/24 edition.

The project is carried out in groups of two or three students and addresses a real-world problem of medium scale. The project comprises several stages:

- identification, delimitation and formulation of the problem to be addressed;
- design, development and evaluation of the solution(s);

- writing of a concise and complete report on the work carried out;
- oral presentation of the developed work.

In particular, the project is presented to students in an open-ended manner, generically identifying the problem to be addressed and the approach to be used. It is the students' responsibility to instantiate this proposal with a concrete project. They are asked to identify:

- the objectives;
- the problem and datasets to be addressed;
- the algorithms to be used;
- the experimental methodology to be adopted;
- the limitations associated with the adopted approaches.

The project descriptions corresponding to the 2023/24 edition can be consulted here. Student involvement in defining their own project acts as a mechanism for motivation and accountability. Together with the oral and public presentation of the achieved results, this aspect contributes to the development of competence **LO5**, identified in Section 4.1.2. The project assessment contributes 60% of the student's final mark.

4.2 Results

This **Curricular Unit**, under the designation **Quantum Data Science**, has been delivered in three editions since the creation of the Master's in Physical Engineering in 2021, resulting from the restructuring of the Integrated Master's in Physical Engineering.

Table 14 presents the number of students enrolled, assessed and approved in the 2021/22, 2022/23 and 2023/24 editions. It is noteworthy that all assessed students achieved a pass. This is due to the project component. In the first two editions this was the only assessment instrument used; in 2023/24 it accounts for 60%. The project is supported by the teaching team throughout the semester, with two intermediate discussion stages and substantial interaction in the theoretical–practical sessions. One consequence of this process has been the approval of students who effectively engage with this component.

Table 15 presents the grades obtained by the students, as well as their distribution across the grading scale, for each academic year. In the first edition, the grades are clearly high (average 17.4 out of 20). This was the first edition of this **CU**, highlighting the need to calibrate certain parameters in the project's assessment.

In subsequent editions, the average grade decreased (14.7 and 15.1), as a result of the corrections applied following the evaluation of results described in the previous paragraph.

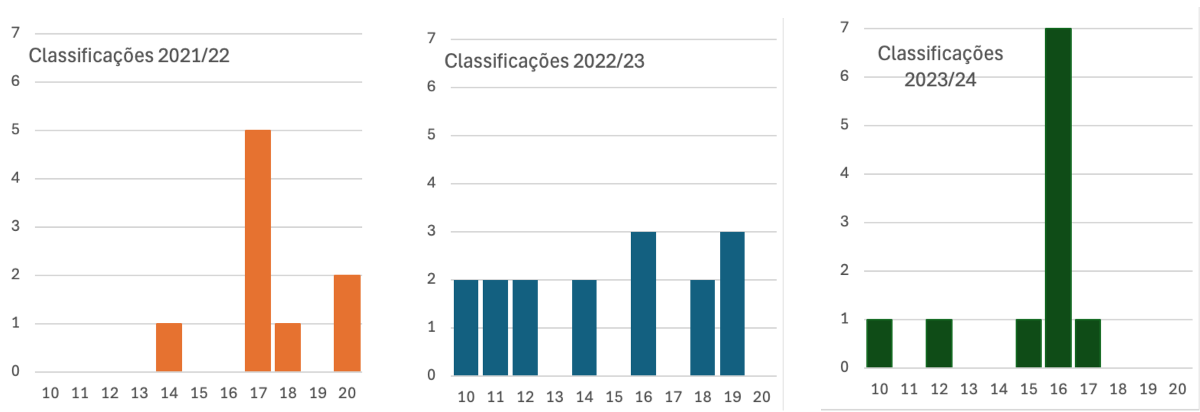
	Enrolled	Assessed	Passed
2021/22	10	9	9
2022/23	19	16	16
2023/24	12	11	11

Table 14: Students in the three editions of the **CU**

The 2023/24 edition was the first to include a written assessment component, with a weight of 40%, implemented through three mini-tests. This innovation was successful, as it required students to prepare for the full range of syllabus contents, and not only those related to the project they selected. There is, however, a peak of grades at 16, resulting from a high uniformity of results in this written component.

	Assessed	[0–9]	10	11	12	13	14	15	16	17	18	19	20	Average
2021/22	9						1			5	1		2	17.4
2022/23	16		2	2	2		2		3		2	3		14.7
2023/24	11		1		1			1	7	1				15.1

Table 15: Students' grades



2021 / 2022 2022 / 2023 2023 / 2024

Figure 17: Student grades

At the University of Minho — within the framework of the **Quality Assurance Internal System – Universidade do Minho (SIGAQ)** — after completing a **Curricular Unit**, students have the opportunity to complete the **Questionnaire for CU assessment - student variant (QUCe)**. This instrument is not unanimously accepted, or at least is not assigned the same meaning, by all agents of the University. Nevertheless, it is a tool that allows the teaching team to form an approximate view of students' perceptions regarding the functioning of the **CU**.

Table 16 presents three questions that the author considered relevant for the evaluation of the **CU** addressed in this report. Students respond on a five-point qualitative scale, ranging from "Poor" to "Excellent".

	2021 / 22	2022 / 23	2023 / 24
Q1 – The organisation of the UC was appropriate	4.2	4.5	4.2
Q3 – The teaching team performed their duties appropriately	4.8	4.2	4.3
Q4 – Your motivation was appropriate given the challenges of the UC	4.3	4.3	4.2

Scale
1 – Very Poor; 2 – Poor; 3 – Fair; 4 – Good; 5 – Excellent

Table 16: Students' responses to the curricular unit evaluation questionnaire

It is easy to observe that students evaluated the functioning of the **CU**, the performance of the teaching team, and their own motivation very positively, with indicators ranging between "Good" and "Excellent". The least positive aspect is the decrease, from 2021/22 to 2022/23, in the perceived quality of the teaching team's performance. At present, there are insufficient data to diagnose the causes of this decrease. It should nevertheless be noted that this evaluation consistently remains within the "Good" to "Excellent" range.

Students from the 2023/24 edition were asked to respond anonymously to a questionnaire designed by the lecturer, concerning their perception of the functioning and relevance of this **Curricular Unit**. The results, corresponding to eight respondents, are presented in Figure 18.

Most students recognise the relevance of **Quantum Machine Learning** in the study programme and consider the assessment methodology to be appropriate. Regarding the sufficiency of prior knowledge in Quantum Computing, they express greater uncertainty, and consider the introduction to classical **Machine Learning** provided at the beginning of this **CU** to be appropriate and sufficient.

In the academic years 2021/22 and 2022/23, a total of 25 students successfully completed this **CU**. Three of these students carried out their dissertations in this area, on the following topics:

- *ADAPT-VQC: Adaptive Variational Quantum Classifier*

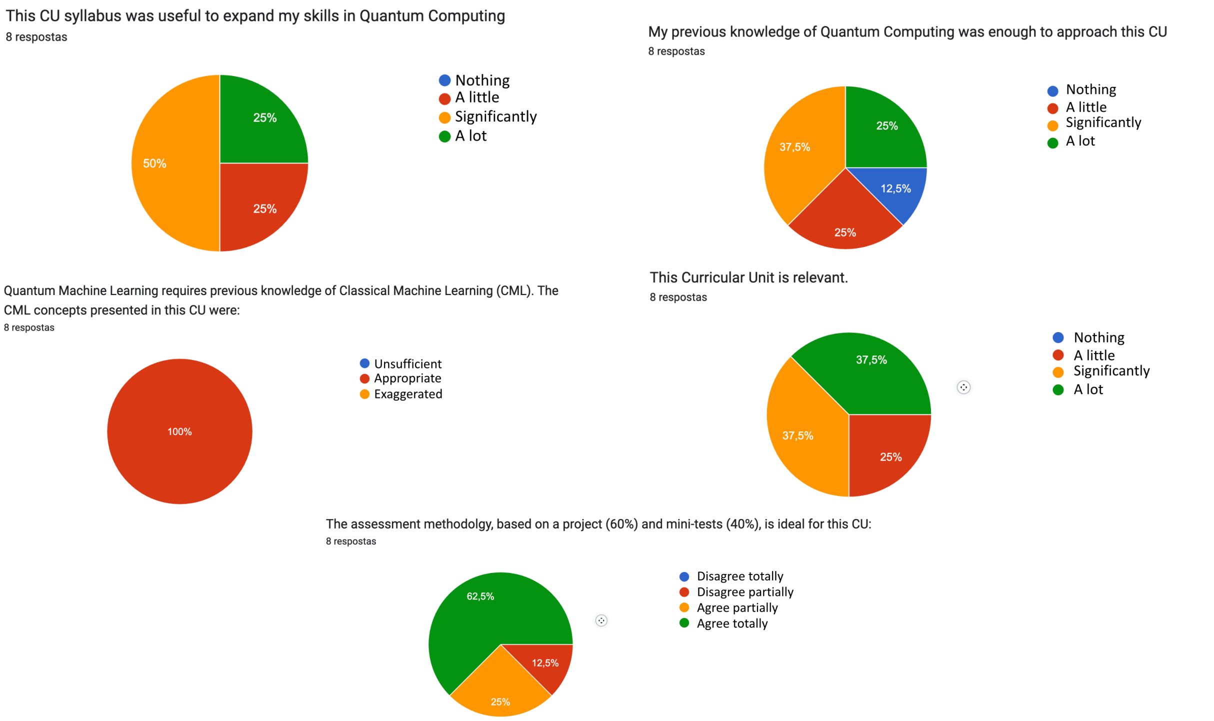


Figure 18: Responses of 2023/24 students to evaluation questions regarding the **CU**

- *Tradeoff between moving targets, gradient magnitude and performance in quantum variational Q-Learning*
- *Data Reuploading and Expressivity / Trainability tradeoff on Policy Gradients*

The outcome of one of these dissertations was published in August 2024 in the journal *Quantum Machine Intelligence* – DOI: 10.1007/s42484-024-00190-z. The former student is currently at Fraunhofer IISB, Germany, as a researcher in the field of *Quantum Machine Learning*. Another student continued to explore the project initiated in this **CU**, with final results currently under submission to a leading journal in the field; the submission is entitled *QAOA for Unit Commitment*. At the time of writing this report, students from the 2023/24 cohort have not yet begun the process of selecting their dissertation topics.

4.3 Summary

This chapter describes the **Curricular Unit Quantum Machine Learning**, offered in the second semester of the **Master in Physical Engineering**, specialisation in **Physics of Information**. Its objectives, syllabus, supporting material and assessment methodology, among other aspects, are established.

The placement of this **CU** within the study plan of this programme makes it clear that students' prior training does not include **Classical Machine Learning**, requiring the allocation of resources (time) to this domain of knowledge. The syllabus related to **QML** is therefore somewhat limited, but this is not considered as very relevant. Indeed, at this level of formal training, it is more important to prioritise critical and autonomous reasoning than to insist on a broad range of topics, which may ultimately result in a degree of superficiality and fragmentation in the internal representation of knowledge constructed by each student.

The main difficulty resulting from the deficit in prior training in **CML** is not the four weeks dedicated to these topics, but rather the fact that students do not have sufficient time to effectively appropriate the associated concepts and competences, nor to establish meaningful connections with other domains of knowledge.

The chapter concludes with an analysis of some results relating to the three editions of this **CU** delivered to date. These results are clearly positive, indicating that students have a favourable perception of the functioning and relevance of the inclusion of **Quantum Machine Learning** in their study plan.

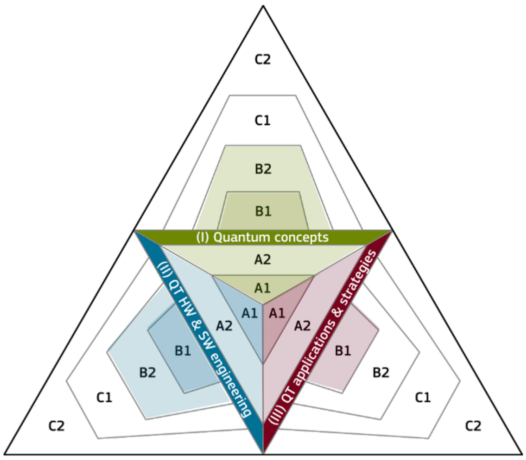
Chapter 5

Final Notes

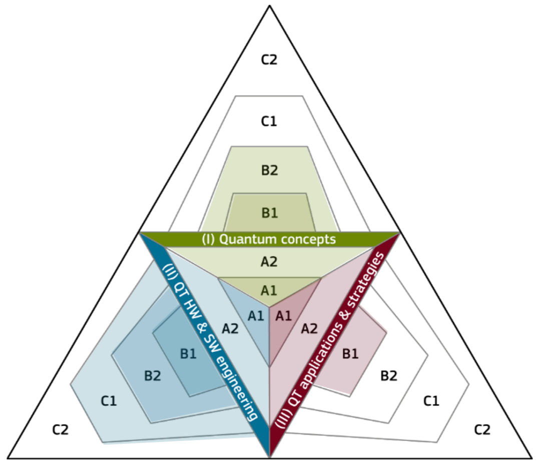
This report presents a detailed description of the **Curricular Unit (CU) Quantum Machine Learning (QML)**, offered in the first year, second semester of the **Master in Physical Engineering**, specialisation in **Physics of Information (PI)**[1]. This educational offering has been properly framed within this Master's study plan and the preceding **BsC in Physical Engineering (LEPhys)**. The specialisation in **PI** places a significant emphasis on the development of competences in the field of Quantum Computing. This **CU** is also appropriately positioned within this domain of knowledge.

The educational offer in Quantum Computing, in general, and in Quantum Machine Learning, in particular, is characterised within the context of the **European Competence Framework in Quantum Technologies** (**ECFQT**). This reference framework constitutes a taxonomy of content, proficiency, and qualification levels in the area of Quantum Technologies. The adoption of a reference taxonomy by the various stakeholders (academia, industry) facilitates the design and comparison of educational projects and competence profiles. This is the primary objective of the **ECFQT** and the reason why this educational offering was characterised using this tool. Figure 19 and Table 17 reproduce the qualification profile and proficiency levels in Quantum Computing of the Master's degree in Physical Engineering, specialisation in Information Physics, at the University of Minho, detailed in Section 3.2.2. These correspond very appropriately to what the **ECFQT** designates as the profile of a Quantum Computing software engineer, whose role typically consists of integrating quantum algorithms into software applications or libraries.

[1] This **CU** is currently designated as **Quantum Data Science**.



(a) Without a dissertation in **QC**



(b) With a dissertation in **QC**

Figure 19: Qualification profile of the Master's degree in Physical Engineering, specialisation in Information Physics, at the University of Minho (see Section 3.2.2)

Without dissertation		**With dissertation**	
I – Quantum concepts			
B2 – Research	**Analyse** problems with quantum methods	B2 – Research	**Analyse** problems with quantum methods
II – h/w and s/w engineering			
B2 – Research	**Analyse** problems with quantum methods	C1 – Specialisation	**Design** the integration of **QC**
III – Applications and strategies			
B1 – Use	**Classify** approaches and applications of **QC**	B1 – Use	**Classify** approaches and applications of **QC**

Table 17: Proficiency levels in Quantum Computing of the Master's degree in Physical Engineering, specialisation in Information Physics, at the University of Minho (see Section 3.2.2)

The European Quantum Readiness Center, within the context of the Quantum Flagship project, promotes the Accord programme, whose objective is to disseminate pioneering initiatives aimed at contributing to the future of the European quantum technologies ecosystem. These initiatives span education, research, and industry. The Accord programme publicises these initiatives on its portal and authorises the respective

Figure 20: Recognition symbol of an Accord initiative.

institutions to use a *label* certifying their competences in the area of quantum technologies—Figure 20. In August 2024, the Quantum Computing education at the University of Minho described in this report was accredited as an Accord educational initiative.

The proposed teaching methodology is based on **participatory** teaching of theoretical concepts and fundamental algorithms and, within the context of theoretical–practical classes, on their development using programming, development, and simulation environments for quantum systems. The emphasis on student participation, particularly in sessions commonly referred to as theoretical, stems from my conviction that student engagement in the teaching–learning process is fundamental not only for the appropriation of content, but above all for the development of competences and the stimulation of curiosity. The use of an **Audience Response System** in some expository sessions is one way of fostering such participation.

The dynamic nature and relative youth of the topic addressed in this **CU** quite naturally raise the question of which syllabus contents should be selected for each edition and of the relevance of certain topics relative to others. It is my conviction that content is not particularly relevant at this level of Higher Education, especially in areas that are constantly being redefined, such as **Quantum Machine Learning**. This excludes, of course, foundational topics such as Quantum Information and Computation, the variational principle, among others.

What is truly relevant are the competences to be developed, which are less sensitive to technological evolution. Higher Education at this level only truly fulfils its mission if the student is the main agent. It is the student's responsibility to cultivate their own autonomy, curiosity, and independence in order to appropriate the relevant concepts and to seek out the questions that genuinely interest them. Once these competences have been developed, they will be well-equipped to face the challenges posed by professional life.

Equally relevant is the role of the lecturer. Acting merely as a transmitter of pre-established syllabus content is clearly inadequate. The lecturer is essentially a facilitator of the aforementioned search undertaken by the student, acting as a guide and motivator throughout this learning and development journey. This vision of teaching is difficult to implement in cohorts with dozens or hundreds of students, as is often the case in the first cycle. However, it is certainly viable in cohorts of modest size, such as those encountered in **Quantum Machine Learning**.

Teaching is also a form of learning. I am convinced that I am today better equipped for this task than in the past. I am confident that in the future I will be even better prepared than I am today.

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Part I
Appendices

Appendix A
Audience Response System

A.1 Classical Machine Learning: Basics

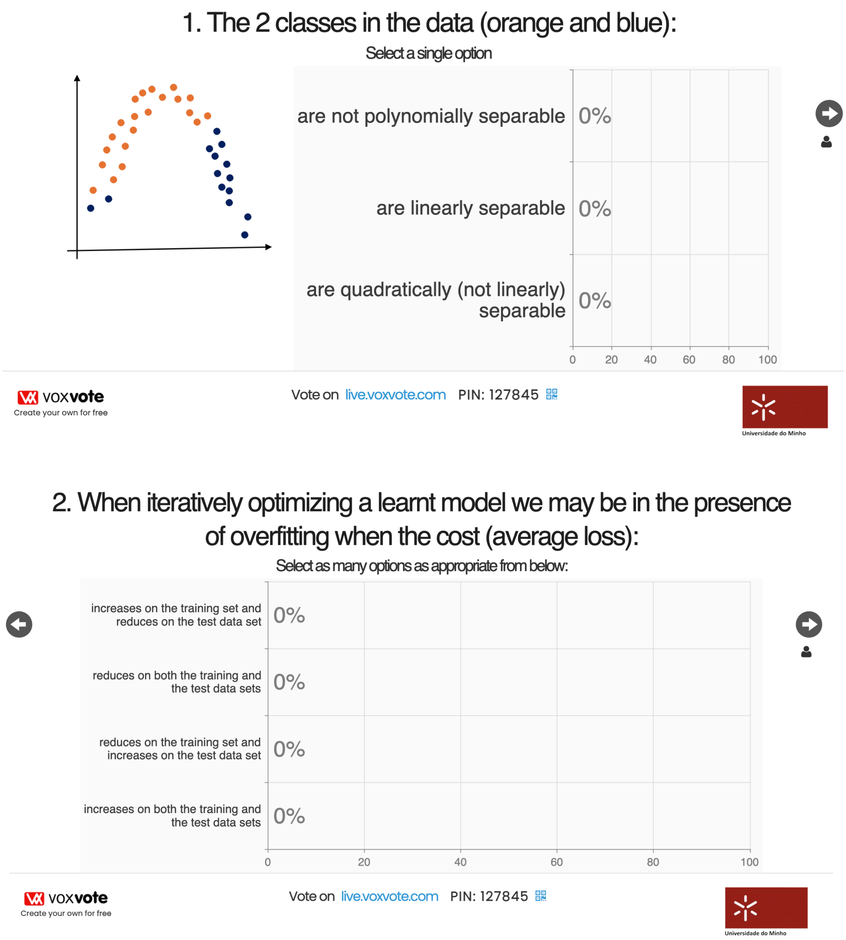
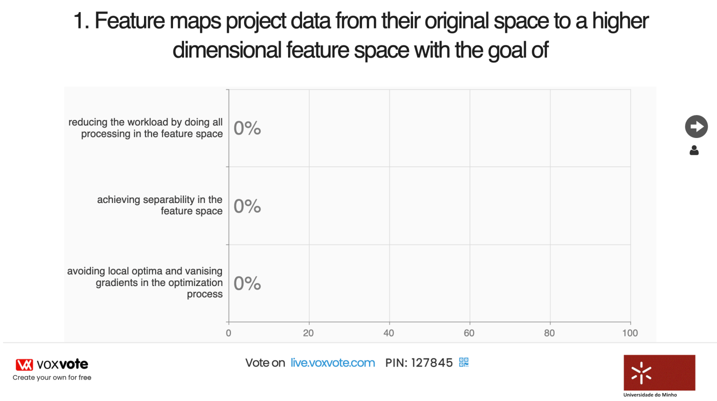
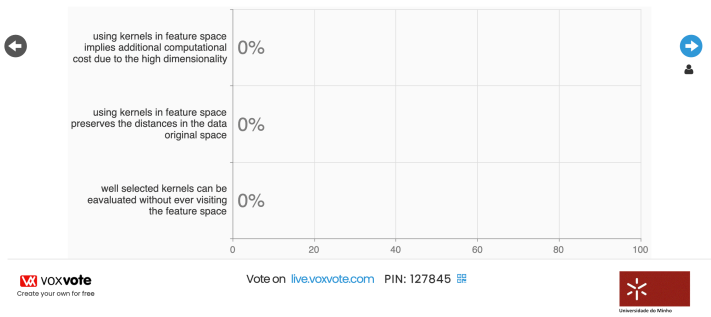
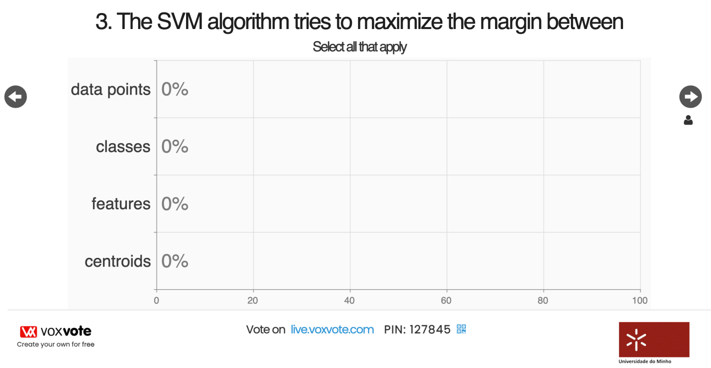


Figure 21: VoxVote - Classical Machine Learning: Basics.

A.2 Kernels and Support Vector Machines







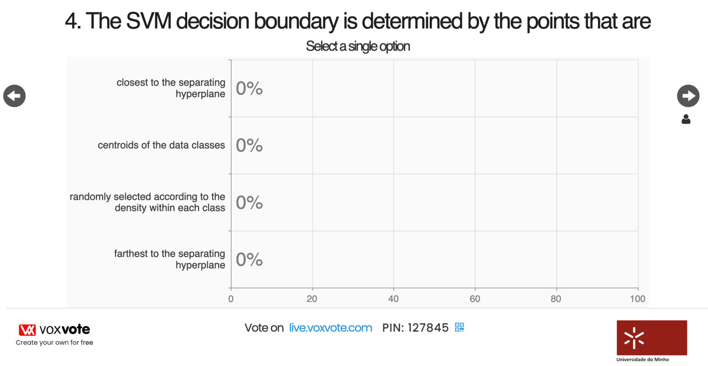


Figure 22: VoxVote - Kernels and Support Vector Machines.

A.3 Data Representation

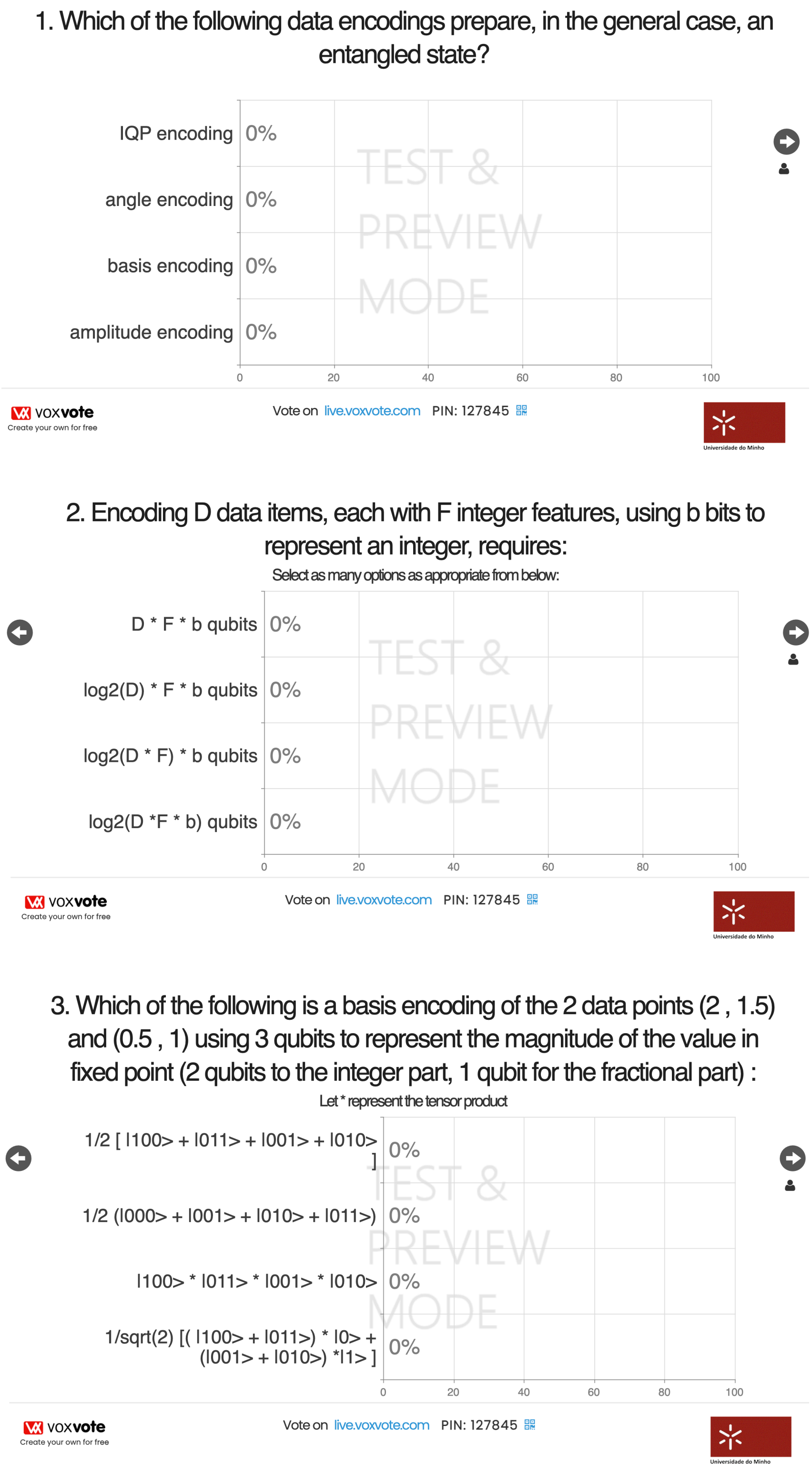
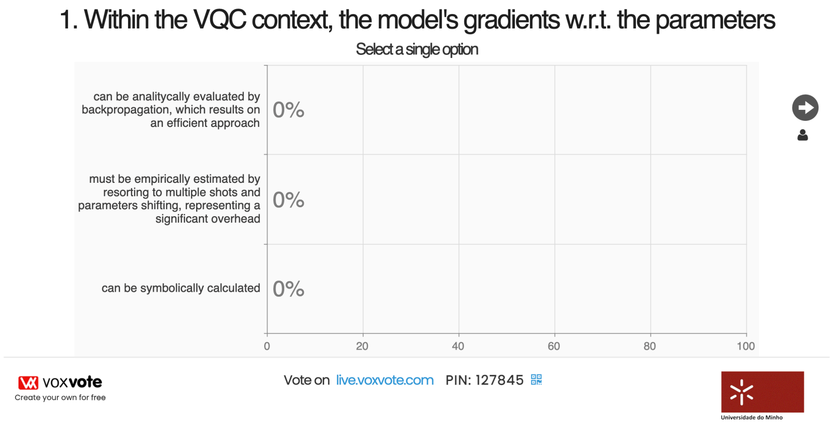
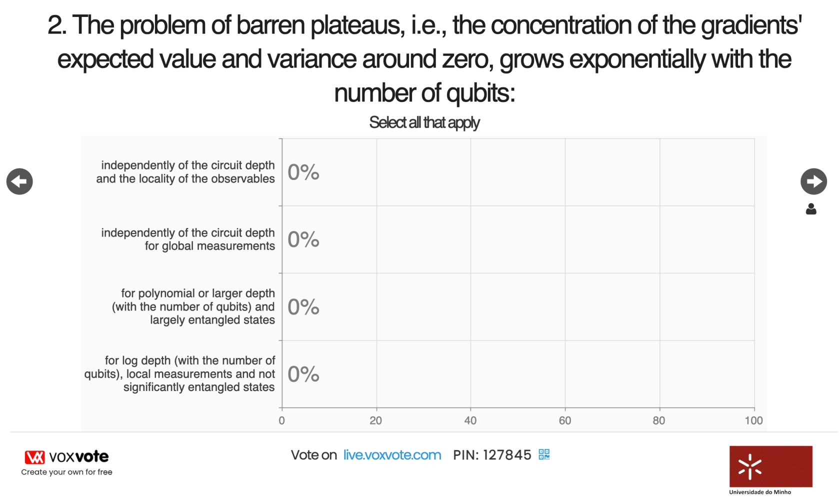


Figure 23: VoxVote - Data Representation.

A.4 Quantum Variational Circuits





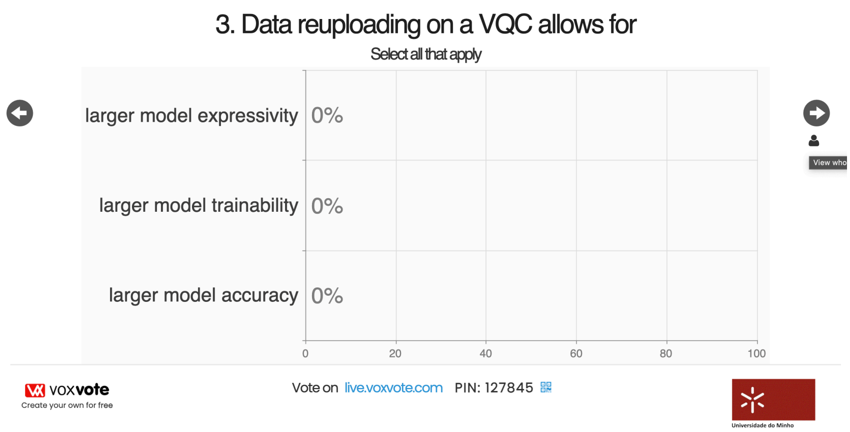
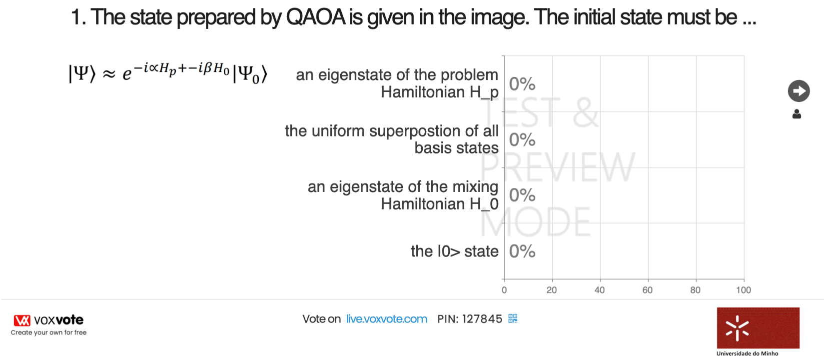


Figure 24: VoxVote - Quantum Variational Circuits.

A.5 Combinatorial Optimization



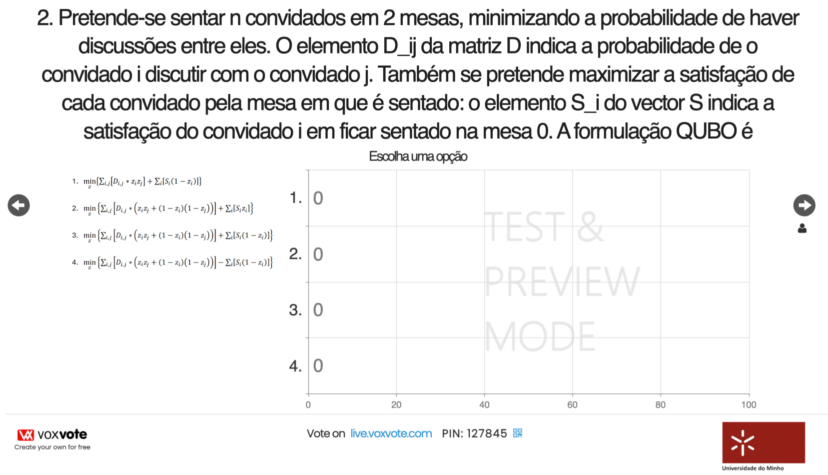


Figure 25: VoxVote - Combinatorial Optimization.

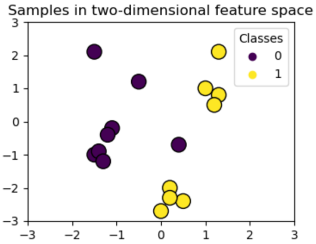
Appendix B
Tests

The following pages present models for the three mini-tests distributed to students in the 2023/24 edition.

Ciência de Dados Quântica | dd.mm.aaa
Mestrado em Engenharia Física – Física da Informação | Mini Teste 1: modelo

1. *[40 %]* - Explain your understanding of over- and under- fitting, in the context of supervised machine learning. Which techniques can be used to mitigate overfitting?

2. *[40 %]* - Consider the following training data set, which will be used to train a Support Vector Machine (SVM):



   a. This data would be better separated using a polynomial feature map of degree 1 or 2? Justify and draw the corresponding separating curve (hyper-plane in the feature space) in the figure.
   b. Mark the support vectors in the figure.
   c. Given your answers above what is the complexity of classifying an unknown data point? Justify.

3. *[20 %]* - The primal model of a support vector machine (SVM) classifies a point x according to the signal of its "distance" to the model' s hyperplane:

$$label(x) = 2 * step(\langle \boldsymbol{\theta}, x \rangle) - 1$$

The dual model, on the other hand, uses the expression below. Describe the rational behind this expression, in terms of the distances that are evaluated.

$$label(x) = 2 * step\left(\sum_{i \in S} \left(\alpha_i y^{(i)} \langle \boldsymbol{x}^{(i)}, x \rangle\right) + b\right) - 1$$

Nome: ____ Número:____________

Figure 26: Model for the first mini test

Ciência de Dados Quântica | dd.mm.aaa
Mestrado em Engenharia Física – Física da Informação | Mini Teste 2: modelo

1. *[40 %]* - Explain, justifying, how many qubits are required to encode on a quantum state in superposition all N data points of the data set, with M features per data point. Consider:
    a. angle encoding;
    b. basis encoding (assume a feature requires τ bits);
    c. amplitude encoding.

2. *[40 %]* - Data reuploading has been proposed as a technique to increase the expressivity of variational quantum circuits. Explain data reuploading and its advantages and disadvantages (with respect to no reuploading).

3. *[20 %]* - The variational quantum algorithm uses a parameterized quantum model:

$$f_\theta(\mathcal{X}) = \langle\psi|S^\dagger(\mathcal{X})\,W^\dagger(\theta)\,O\,W(\theta)\,S(\mathcal{X})|0\rangle$$

to approximate a machine learning problem $f^*: \mathcal{X} \longrightarrow \mathcal{Y}$. Explain the role of θ and the process by which the quantum model can progress towards a good approximation of f^*.

Nome: ____________________ Número: ____________

Figure 27: Model for the second mini test

1. *[40 %]* - QAOA's ideal unitary is given by: $\boldsymbol{e^{-i(\beta H_0+\alpha H_p)}}$. Explain what is represented by $\boldsymbol{H_0}$, $\boldsymbol{H_p}$, $\boldsymbol{\beta}$, and $\boldsymbol{\alpha}$.

2. *[40 %]* - Consider a training data set $\mathcal{D} = \{x^{(1)}, \cdots, x^{(D)}\}$ with D data points. Each data point has 3 features: $x^{(i)} = \left(x_1^{(i)}, x_2^{(i)}, x_3^{(i)}\right)$.

   Draw the circuit that would allow you to compute an estimate of $\kappa\left(x^{(1)}, x^{(2)}\right)$ using the Inversion Test and angle encoding, i.e., $\phi\left(x_1^{(i)}, x_2^{(i)}, x_3^{(i)}\right) = R_Y\left(x_1^{(i)}\right) \otimes R_Y\left(x_2^{(i)}\right) \otimes R_Y\left(x_3^{(i)}\right)$

   Explain also how you would compute the estimate $\tilde{k}\left(x^{(1)}, x^{(2)}\right)$ from multiple measurements of the proposed circuit.

3. *[20 %]* - For a quantum kernel to have any possibility of achieving a quantum advantage it must be computationally hard to evaluate (or estimate) classically. To achieve that computational hardness, it is often suggested that the feature map associated with the kernel must have multiple layers (e.g., IQP with more than 1 layer or multiple layers of product states (rotations) followed by entangling gates). Comment on the reasons why the following statement is not true:

   "*The more layers associated with the kernel feature map (or data encoding), the more probable is the quantum advantage and better classification results will be obtained.*"

Nome: ______________________________ Número:____________

Figure 28: Model for the third mini test

Luís Paulo Peixoto dos Santos

January 2026